

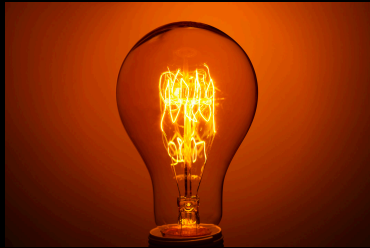
Cost of Computation

PhD Proposal

Lokesh Boominathan

Advisor: Dr. Xaq Pitkow

Motivation



20 Watts



~ 20 Watts



ALPHAGO

~ 1 Megawatt
(50,000 brains!)

How the brain handles the different computation costs is crucial in designing highly energy-efficient real-world systems.

Brains have evolved over millions of years to be highly energy-efficient.

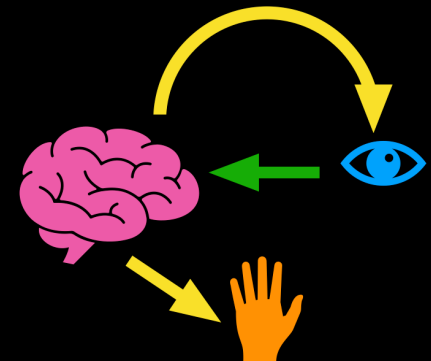
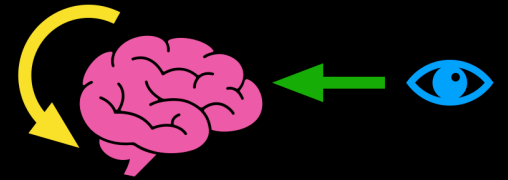
Hsu, Jeremy. "How IBM got brainlike efficiency from the TrueNorth chip." *IEEE Spectrum* 51.10 (2014): 17-19.

Briman, E. "Another look at alphago vs. Lee Sedol: The power angle." online, Accessed (2018): 12-04

Introduction

Brain computations under noise and energy constraints?

- Part I: Inference under resource constraints
 - An **abstract** mathematical model to study how brain controls **internal feedback**
- Part II: Inference and control under resource constraints
 - Applying such models to **actual brain data** to study how it controls **external feedback** - attention and action choices



Outline

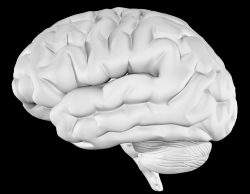
Problem
formulation



Solution &
interpretation



Biological
relevance



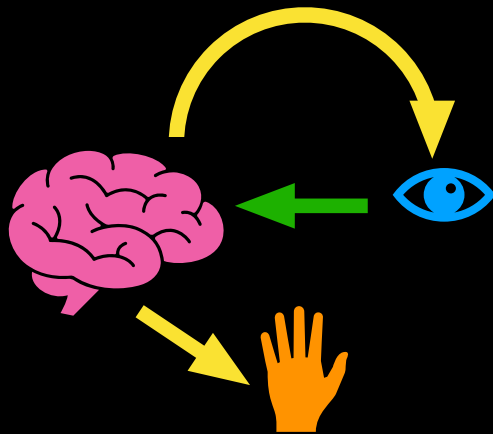
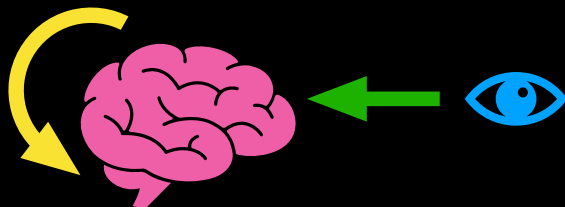
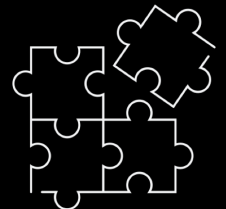
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Solution &
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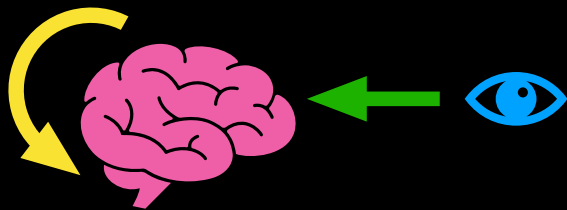


Toy problem



Outline

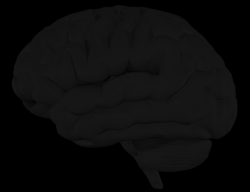
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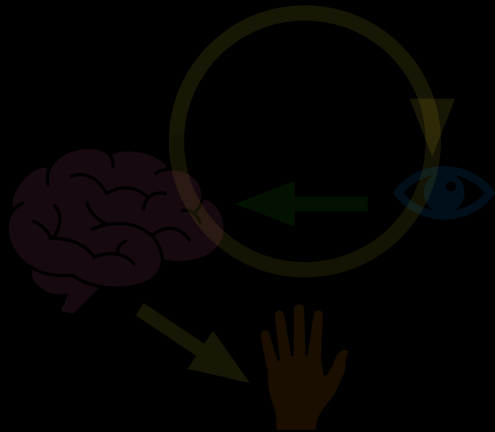
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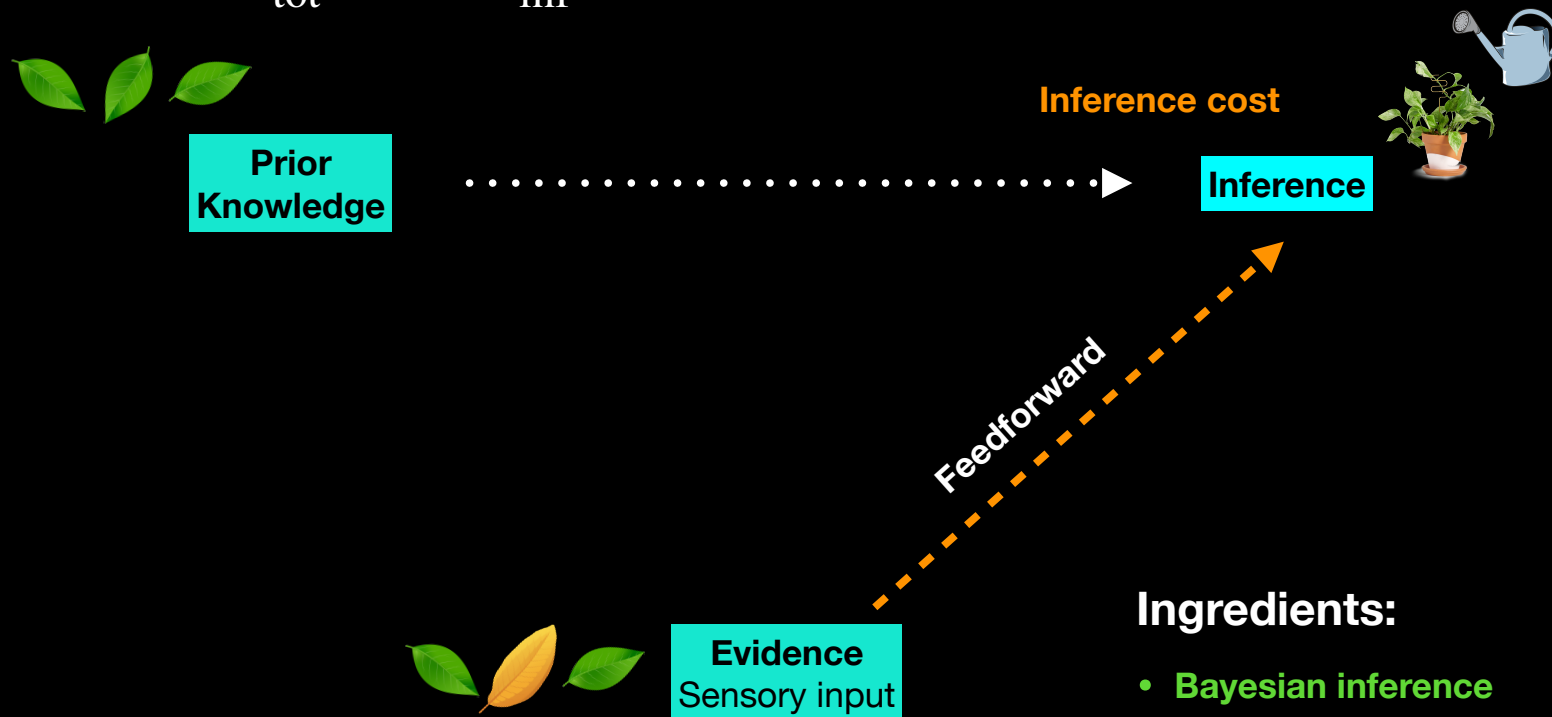


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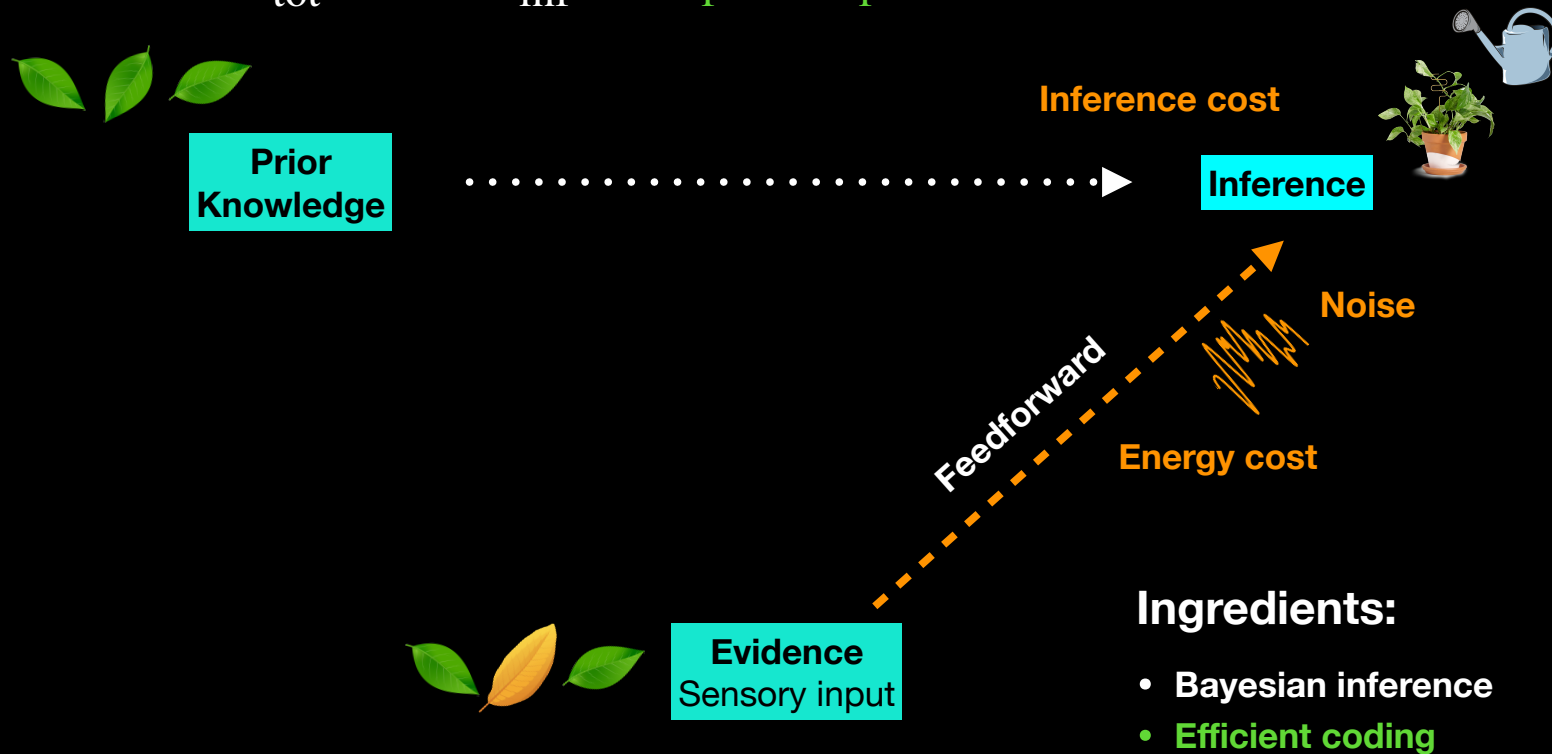
Bayesian inference combines evidence with prior

$$\text{Cost}_{\text{tot}} = \text{Cost}_{\text{inf}}$$



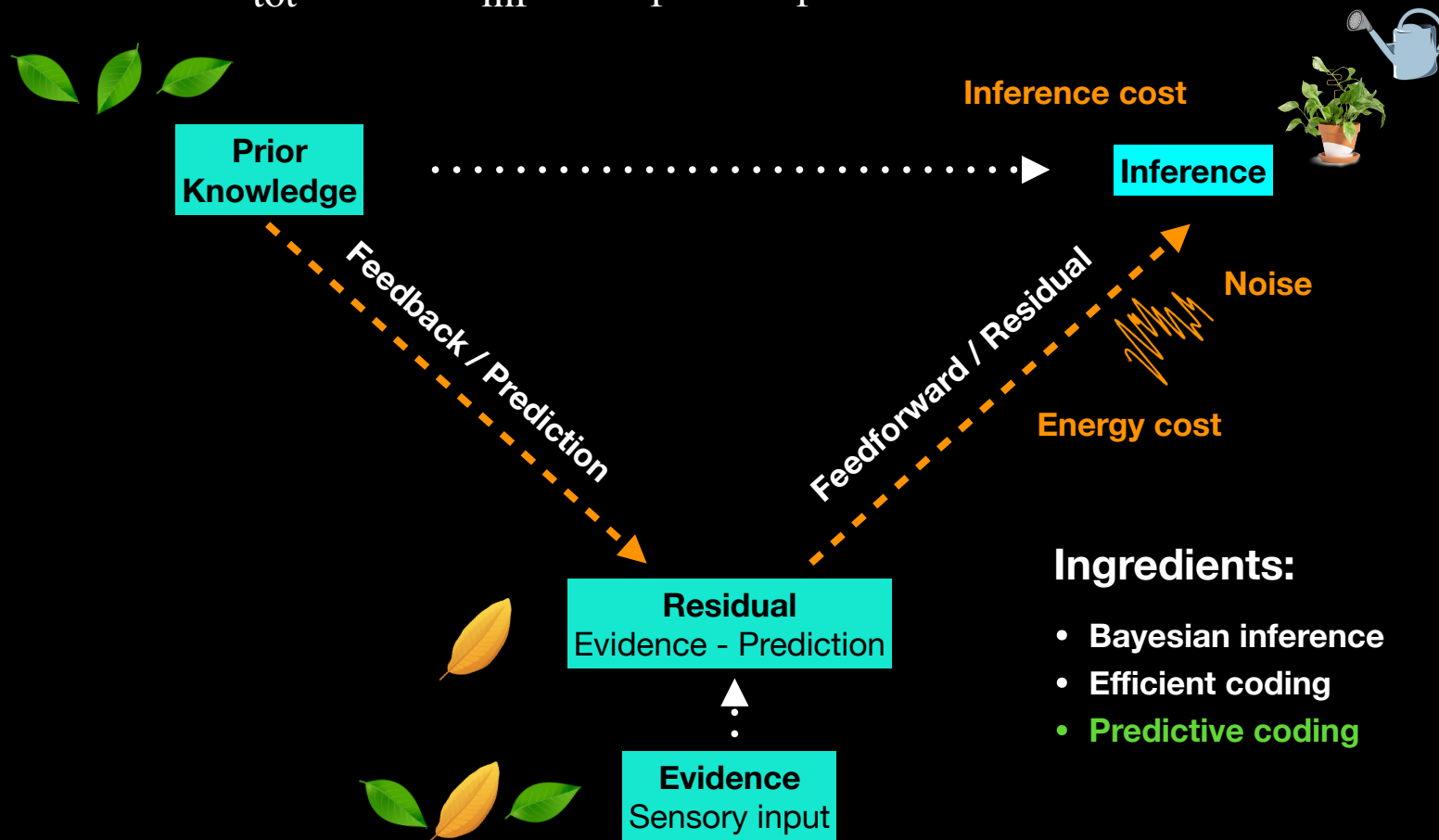
Efficient coding aims efficient resource utilization

$$\text{Cost}_{\text{tot}} = \text{Cost}_{\text{inf}} + w_f \text{Cost}_f$$



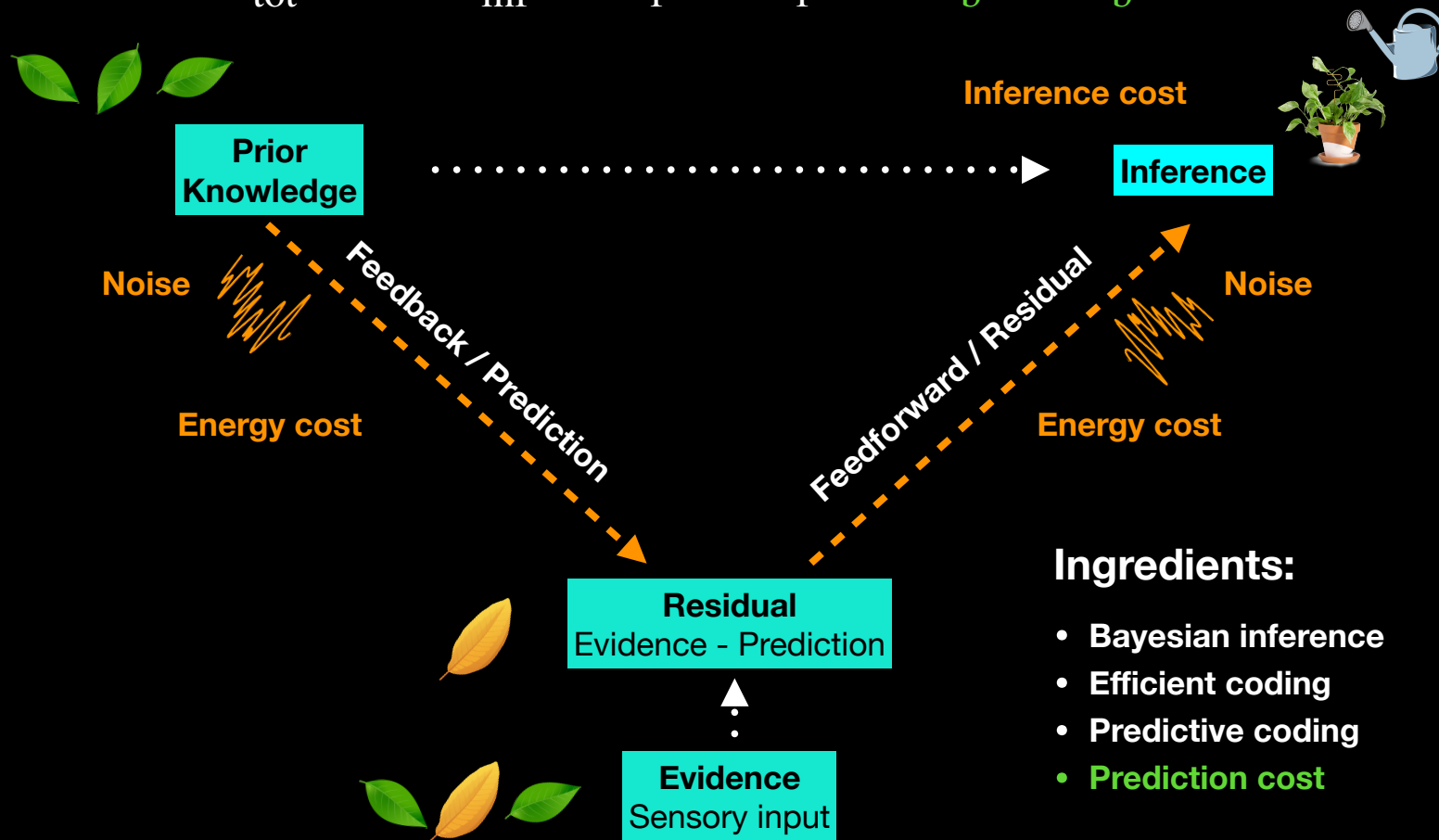
Predictive coding uses feedback to reduce feedforward energy

$$\text{Cost}_{\text{tot}} = \text{Cost}_{\text{inf}} + w_f \text{Cost}_f$$



But predictions come with a cost too!

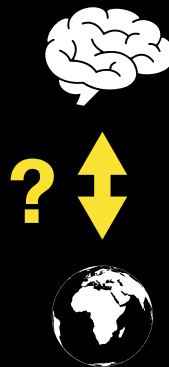
$$\text{Cost}_{\text{tot}} = \text{Cost}_{\text{inf}} + w_f \text{Cost}_f + w_b \text{Cost}_b$$



Our framework includes all costs and noise

$$\text{Cost}_{\text{tot}} = \text{Cost}_{\text{inf}} + w_f \text{Cost}_f + w_b \text{Cost}_b$$

Inference
Cost



Feedforward
Cost



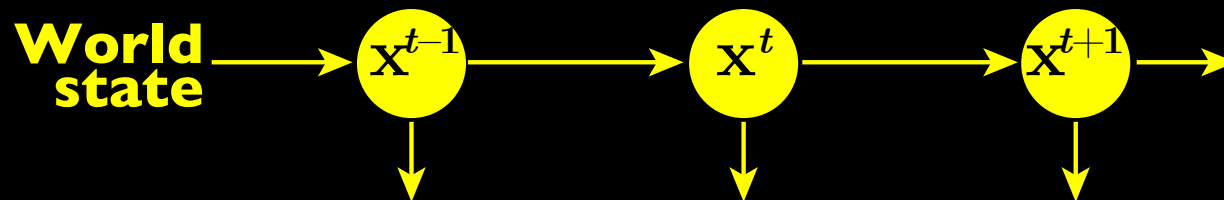
Feedback
Cost



w_b , w_f : Feedback, feedforward weight

Inference problem: Hidden Markov Model

Specifically: Linear dynamics with Gaussian noise

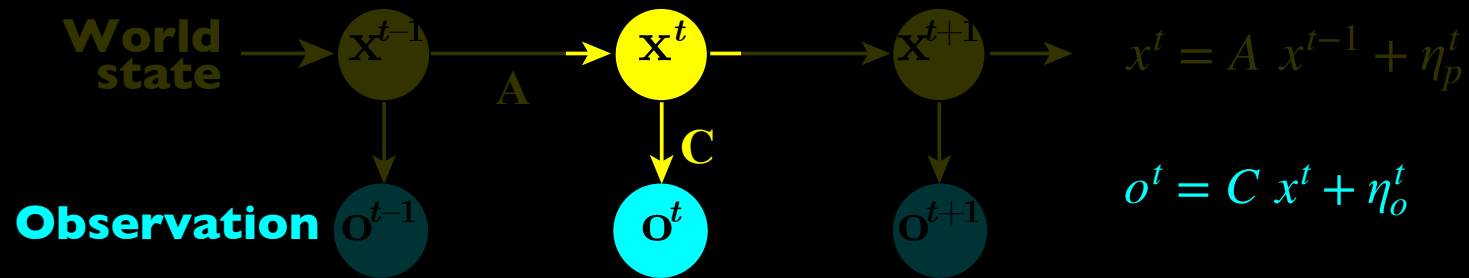


No infrastructure and cost for sending predictions

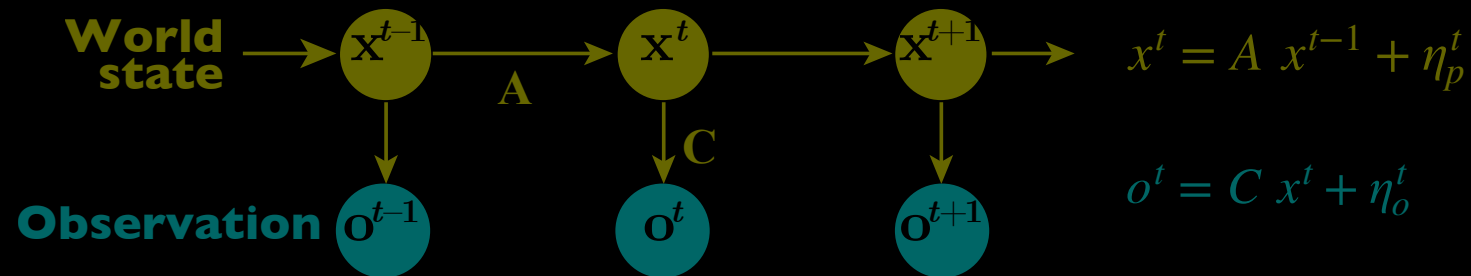
World transitions in a structured way



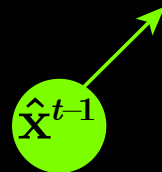
Sensory observations of the world is noisy



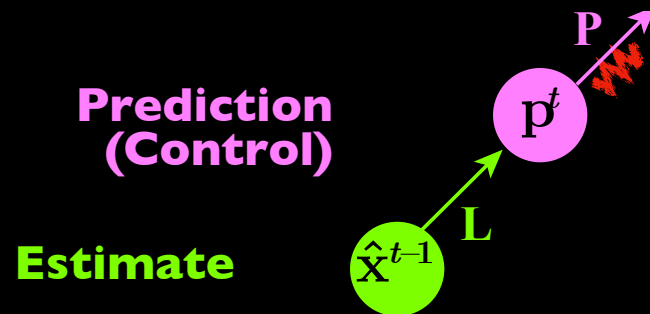
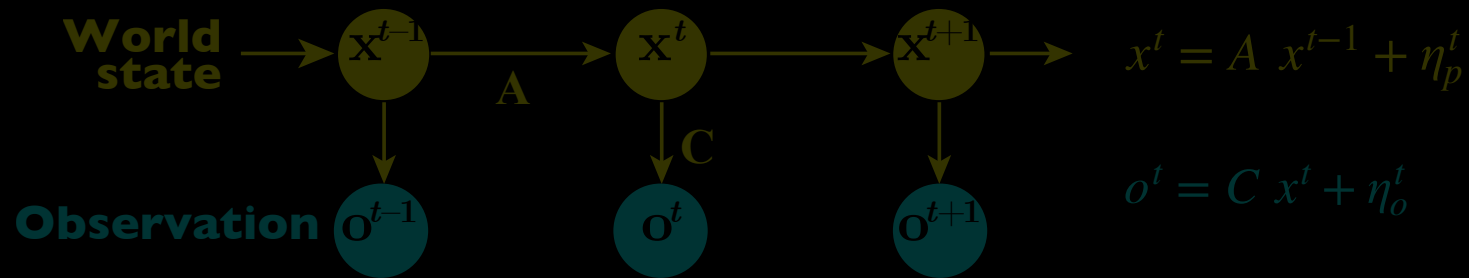
Brain has a prior knowledge about the world



Estimate



Prior based prediction is sent to the sensory level

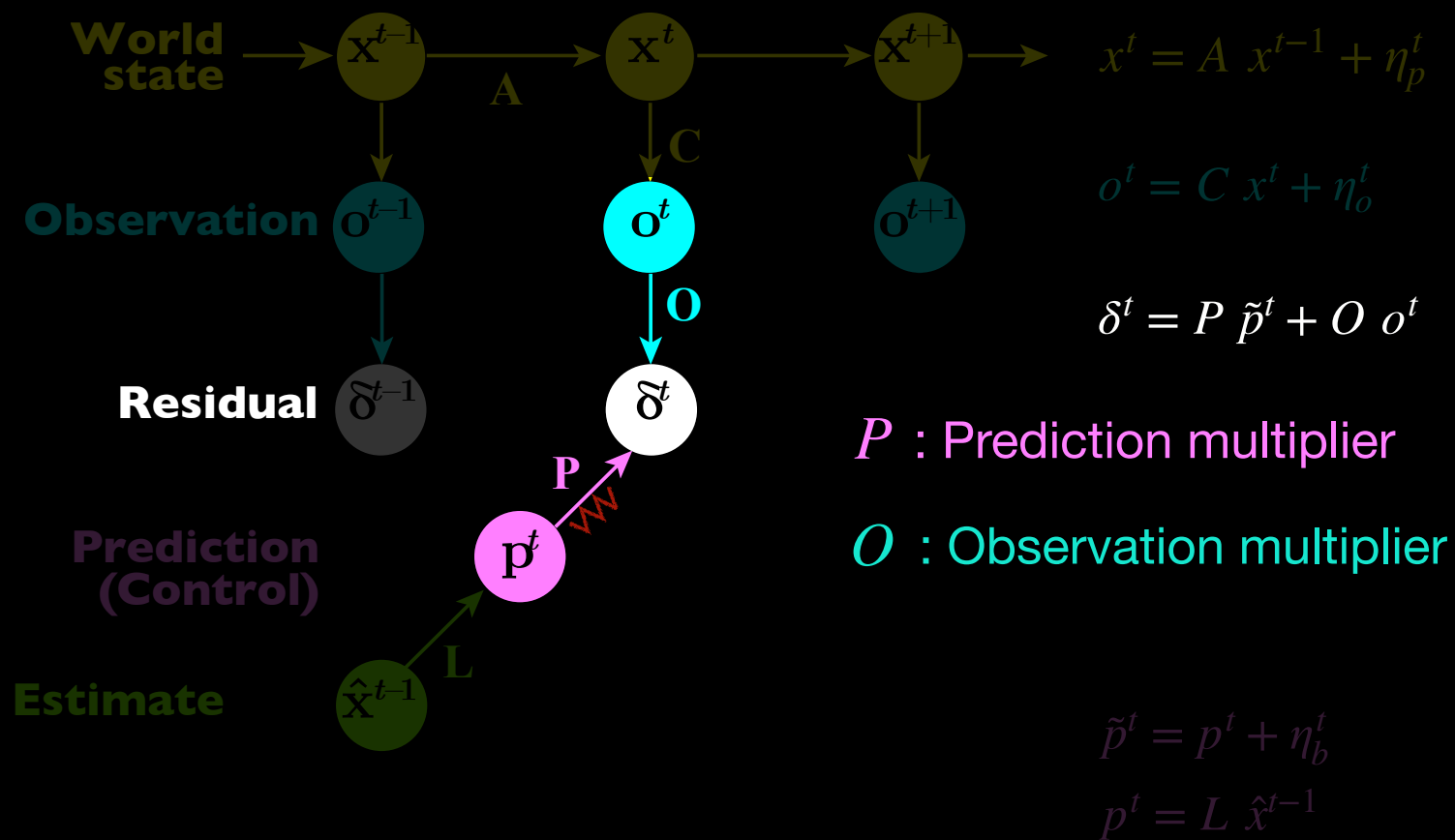


L : Control Gain

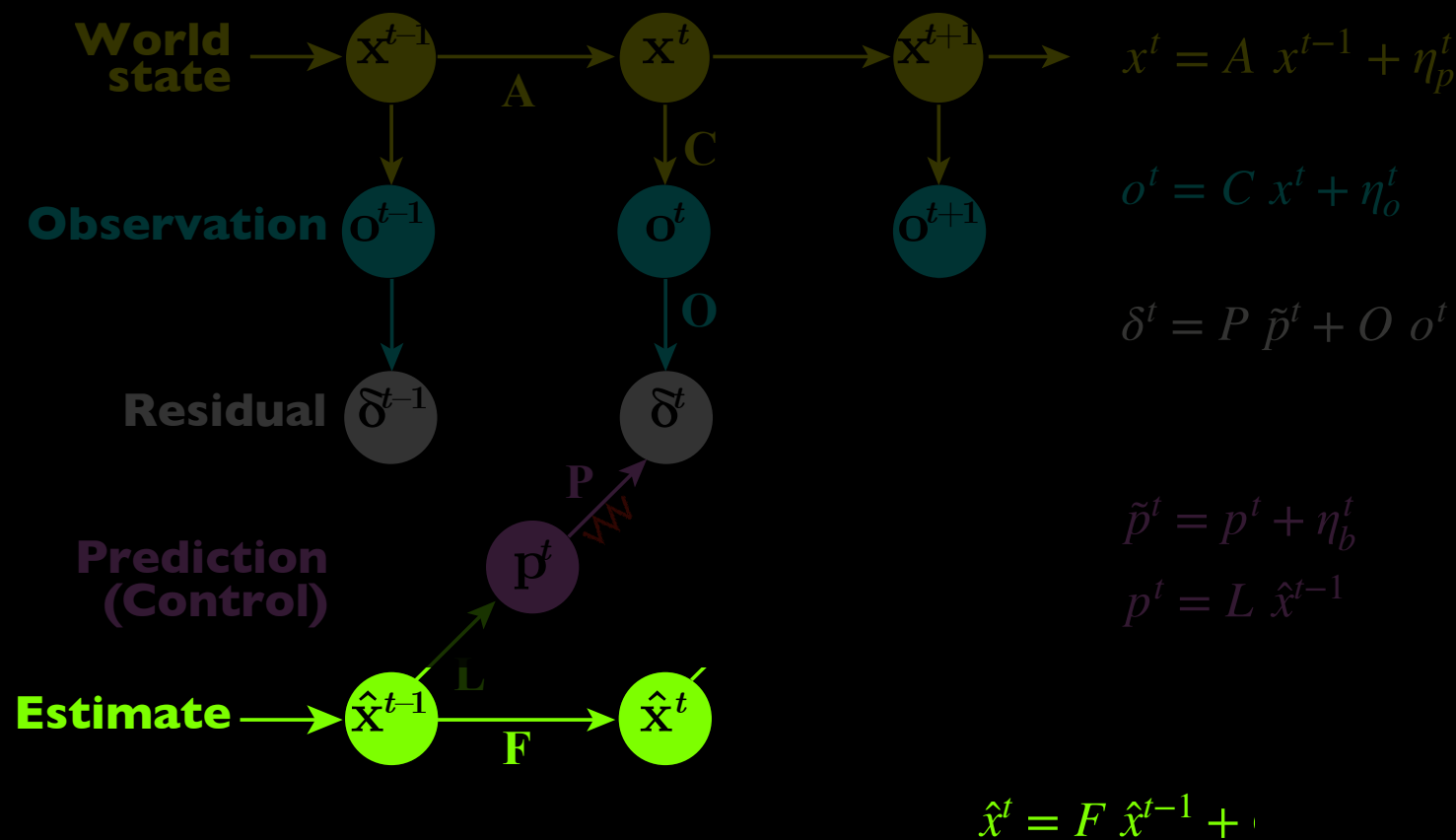
$$\tilde{p}^t = p^t + \eta_b^t$$

$$p^t = L \hat{x}^{t-1}$$

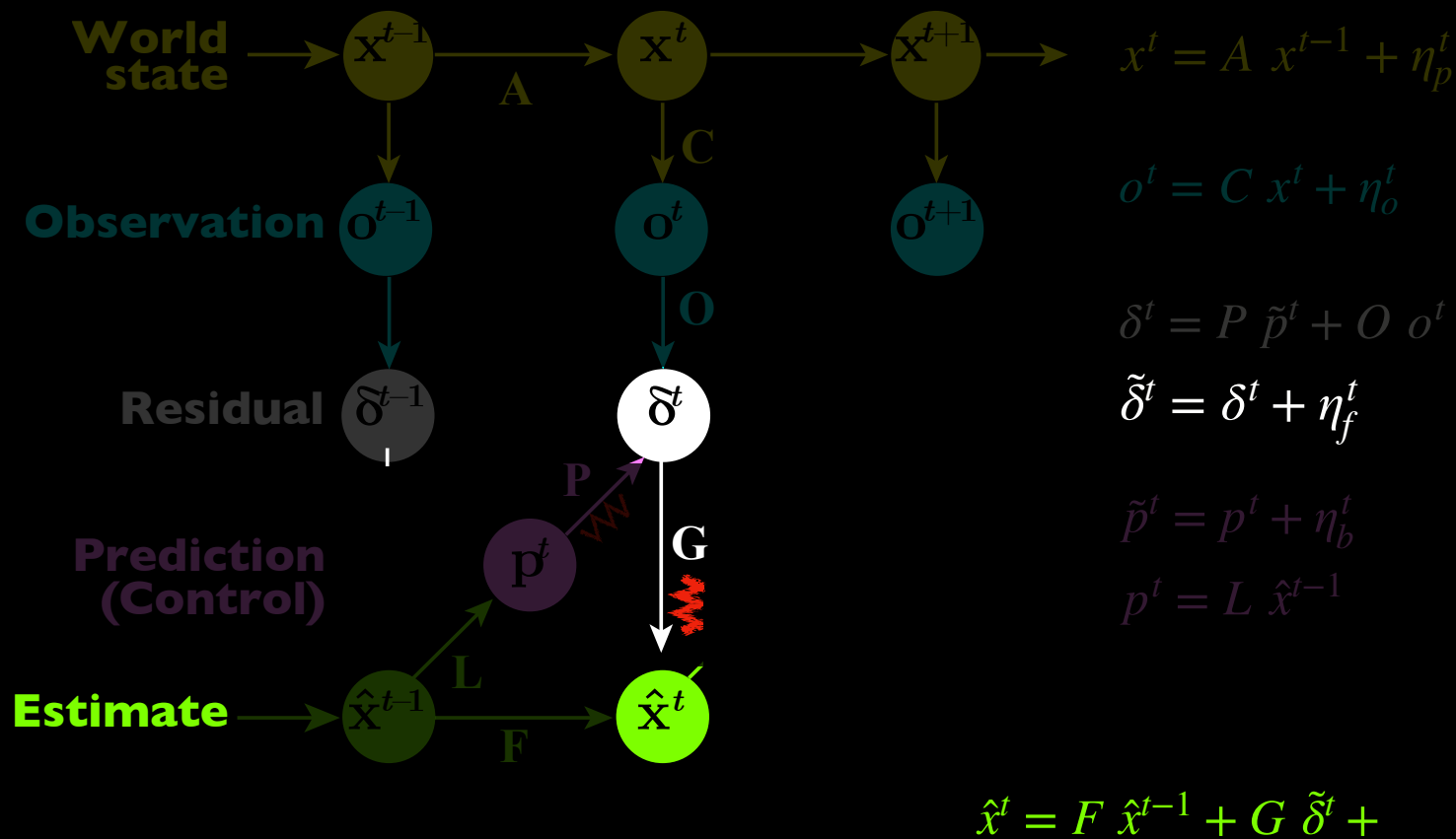
Residual is based on prediction and observation



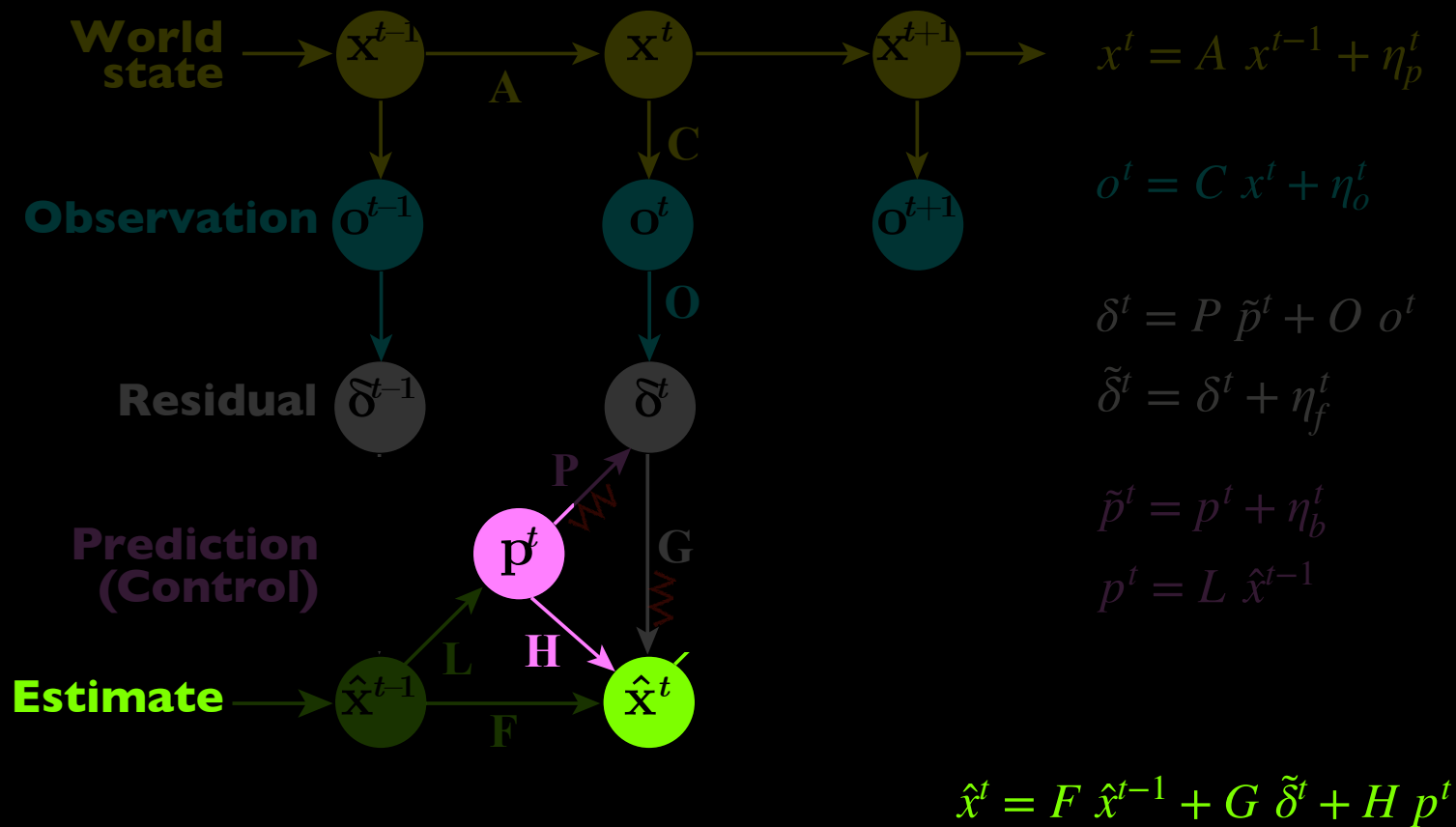
Previous estimate is used to update the current



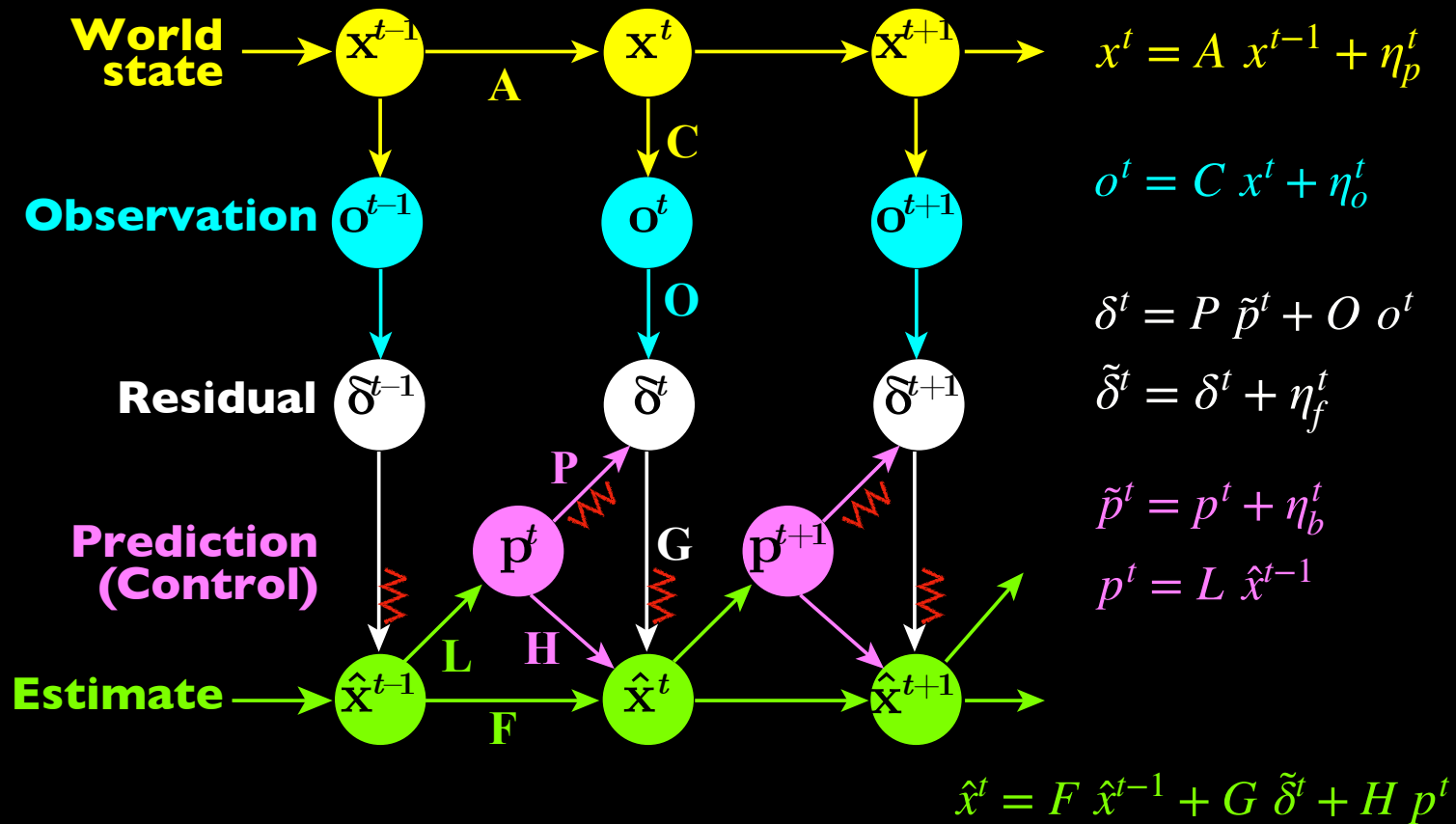
Noisy residual is used to update the estimate



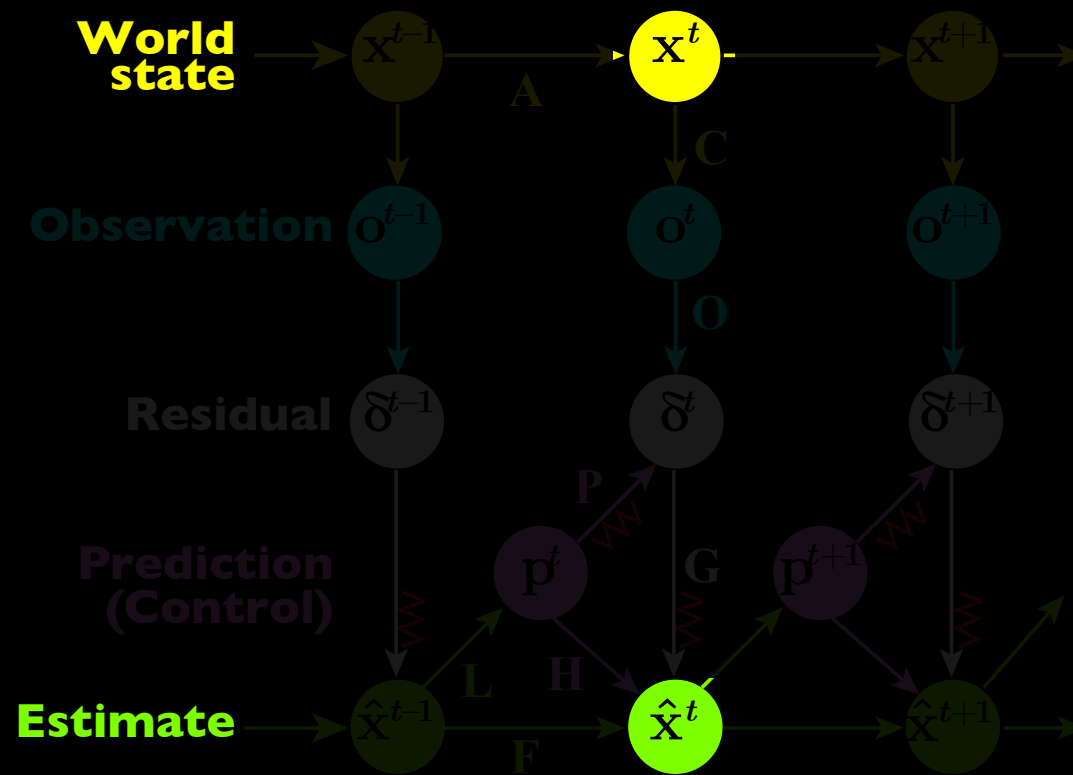
Sent prediction is used to update the estimate



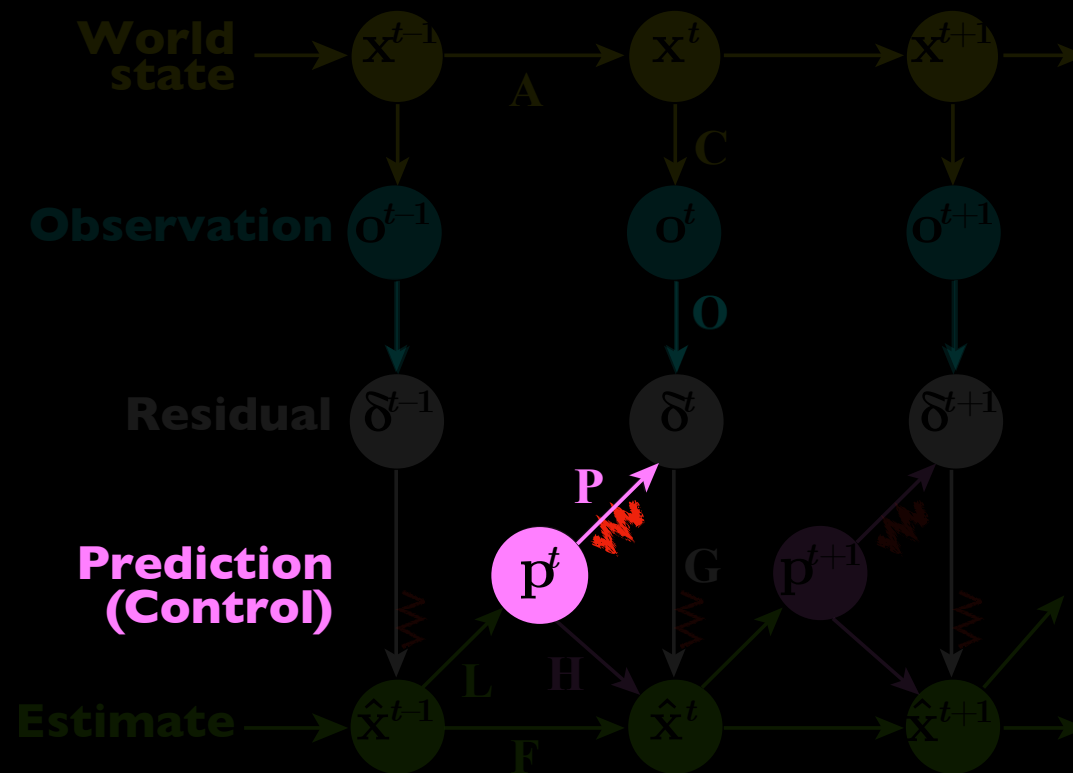
Infrastructure for noisy predictions and residuals



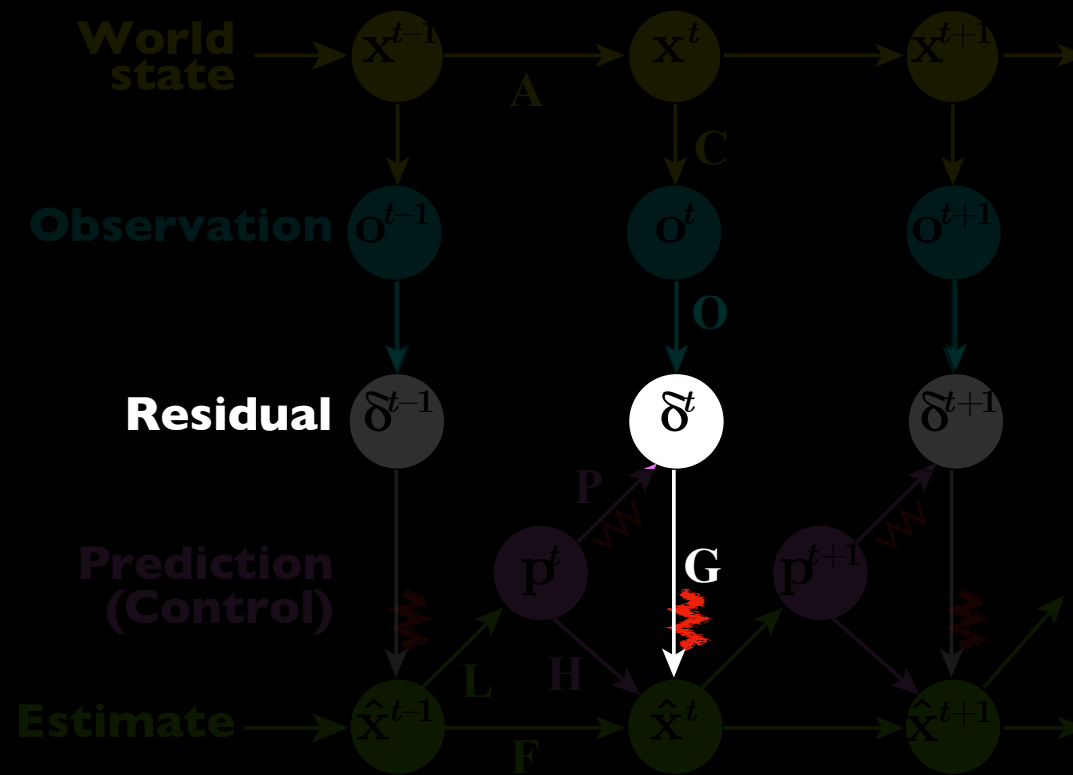
$$\text{Cost}_{\text{tot}} = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \left(\underbrace{\|x^t - \hat{x}^t\|^2}_{\text{Cost}_{\text{inf}}} + j \right)$$



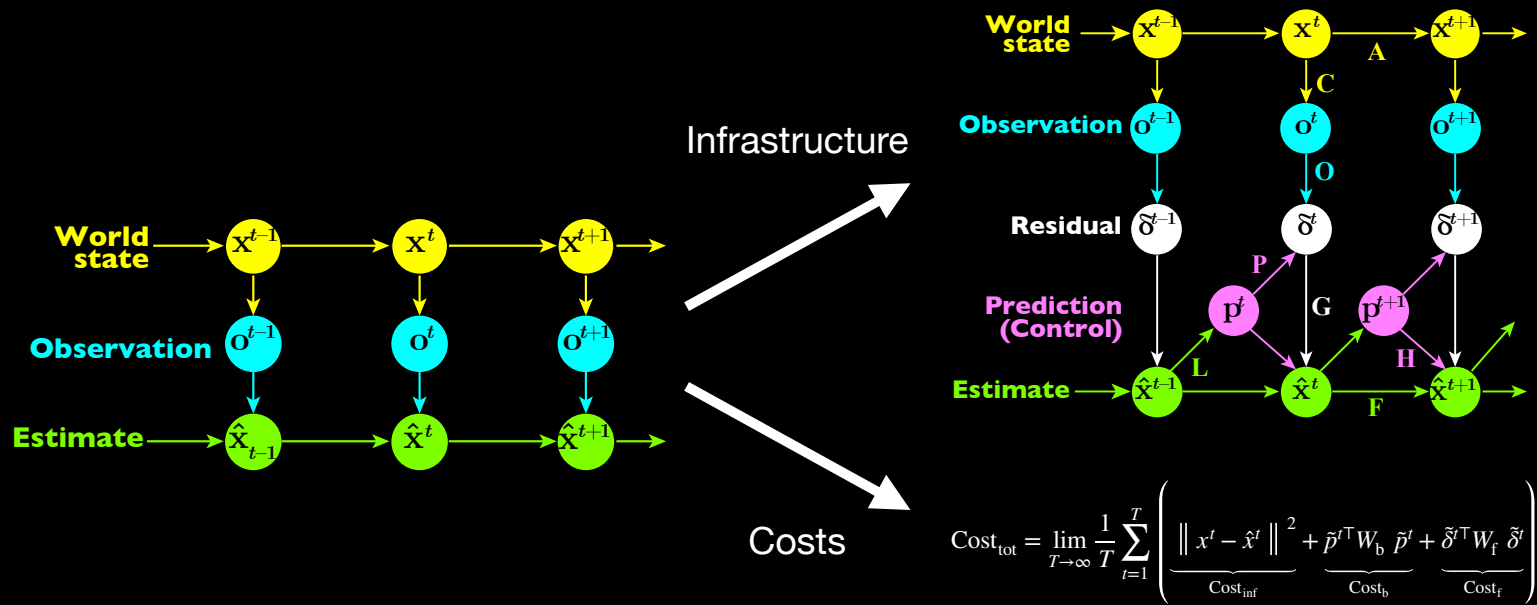
$$\text{Cost}_{\text{tot}} = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \left(\underbrace{\|x^t - \hat{x}^t\|^2}_{\text{Cost}_{\text{inf}}} + \underbrace{\tilde{p}^{t\top} W_b \tilde{p}^t}_{\text{Cost}_b} + \right)$$



$$\text{Cost}_{\text{tot}} = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \left(\underbrace{\|x^t - \hat{x}^t\|^2}_{\text{Cost}_{\text{inf}}} + \underbrace{\tilde{p}^{t\top} W_b \tilde{p}^t}_{\text{Cost}_b} + \underbrace{\tilde{\delta}^{t\top} W_f \tilde{\delta}^t}_{\text{Cost}_f} \right)$$



Expanded to incorporate costs and infrastructure for predictions



Given: World, sensory, and channel properties.

Goal: Optimal inference update mechanism.

$$\arg \min_{L, P, O, F, G, H} \text{Cost}_{\text{tot}}$$

L : How much is prediction amplified?

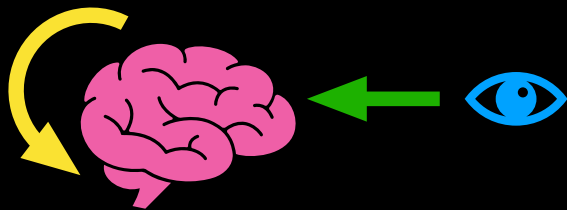
P : How prediction forms residual?

O : How observation forms residual?

F, G, H : How brain updates inference?

Outline

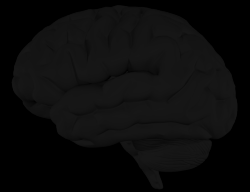
Problem
formulation



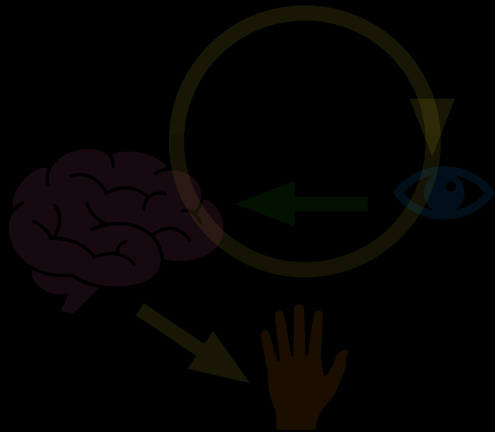
Solution &
interpretation



Biological
relevance



Problem
formulation



Solution &
interpretation



Toy problem



Outline

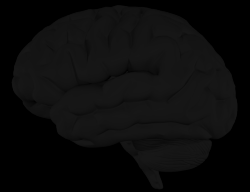
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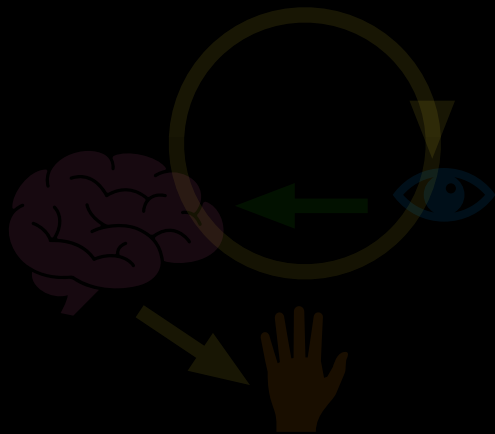
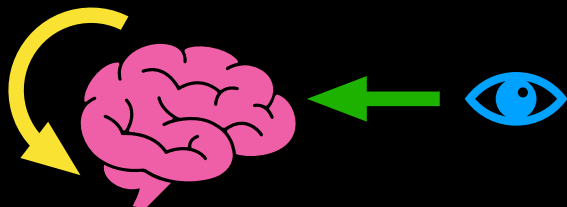
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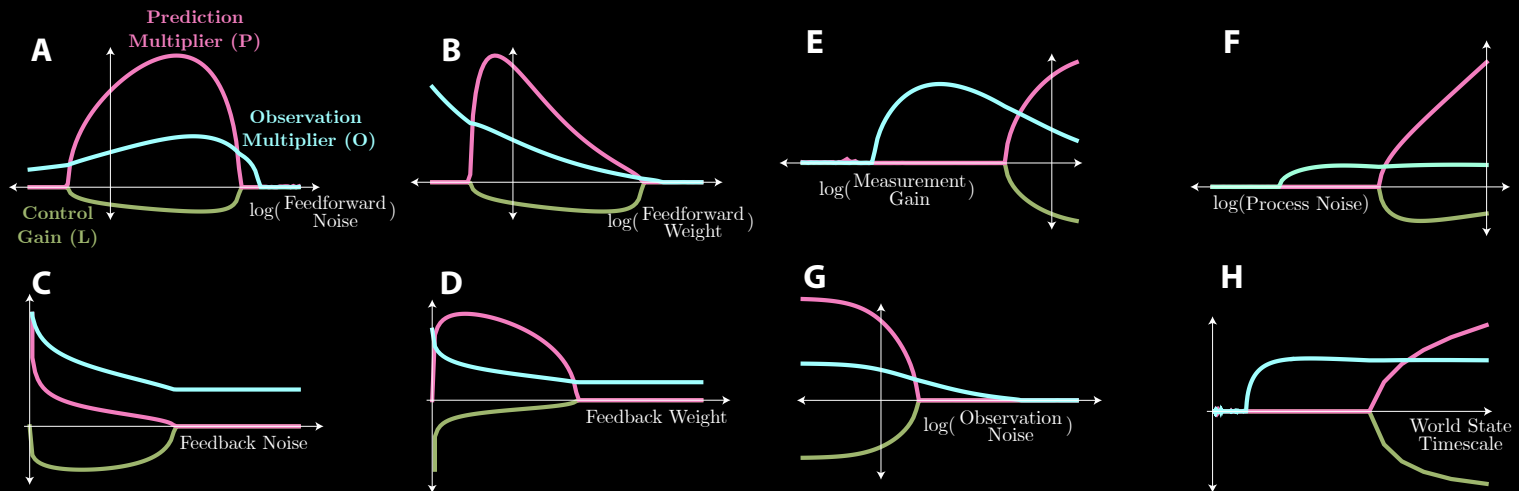
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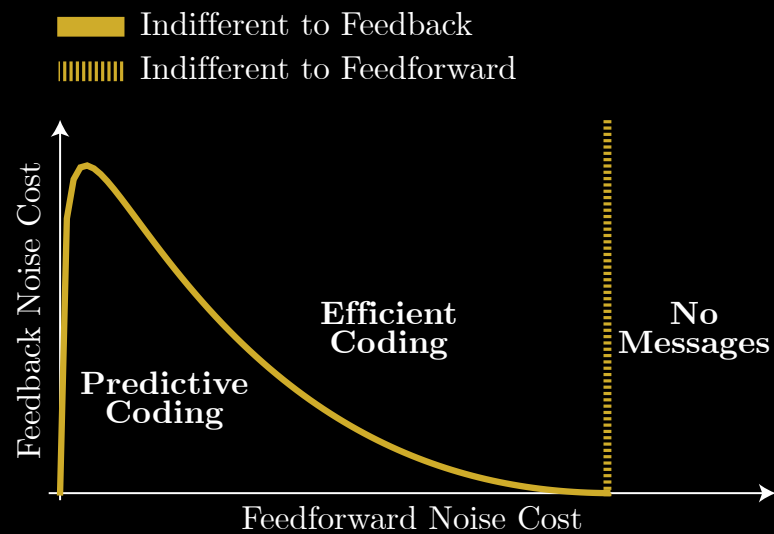
Toy problem



Optimal strategy changes with constraints

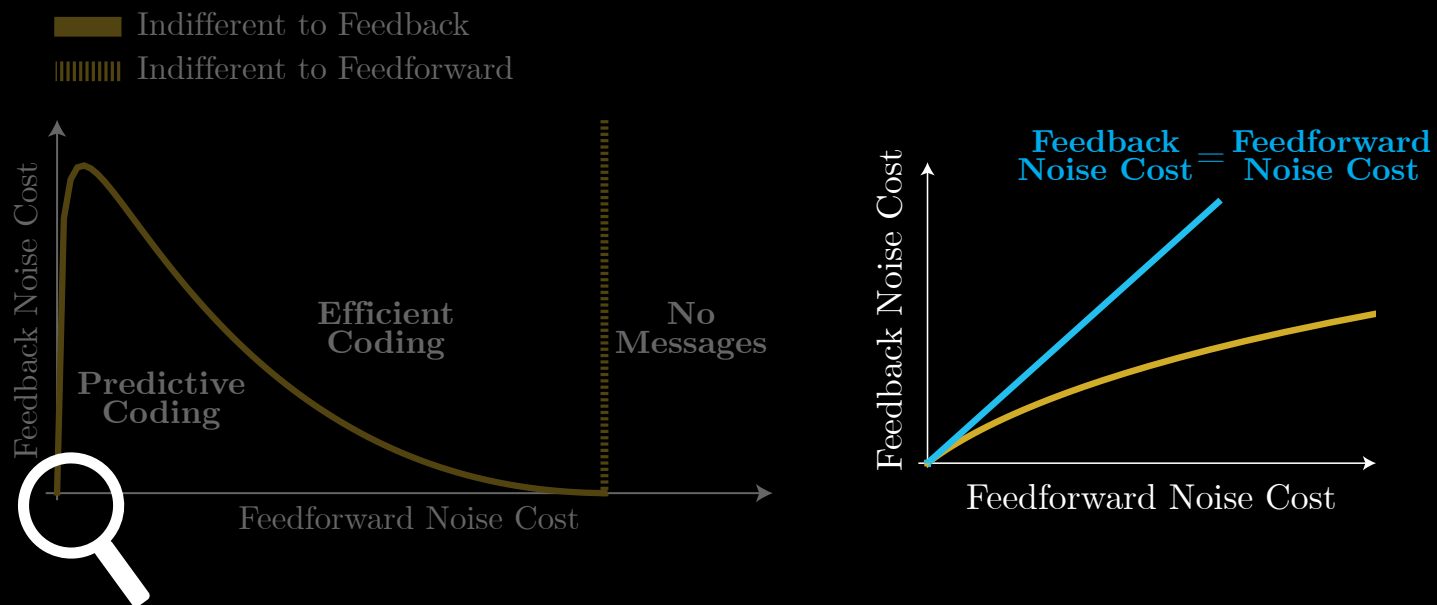


Feedback usefulness condition is more stringent



When feedback is useful, feedforward is definitely useful

Feedback usefulness condition is more stringent

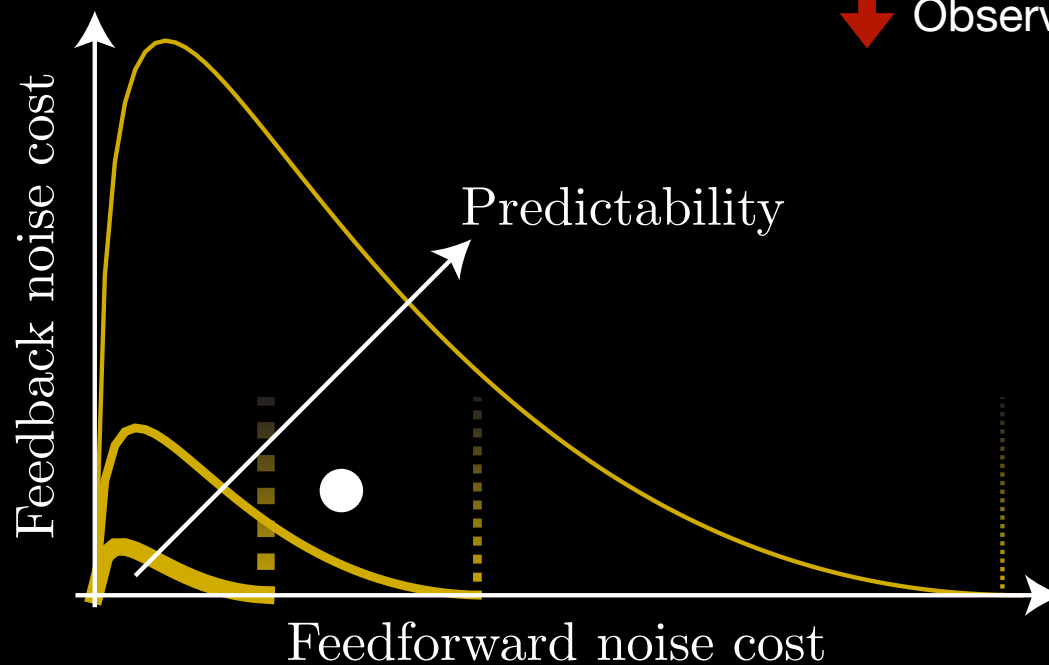


Prediction useful only when feedback channel is way cheaper/noise-free than the feedforward

High predictability conducive for predictions

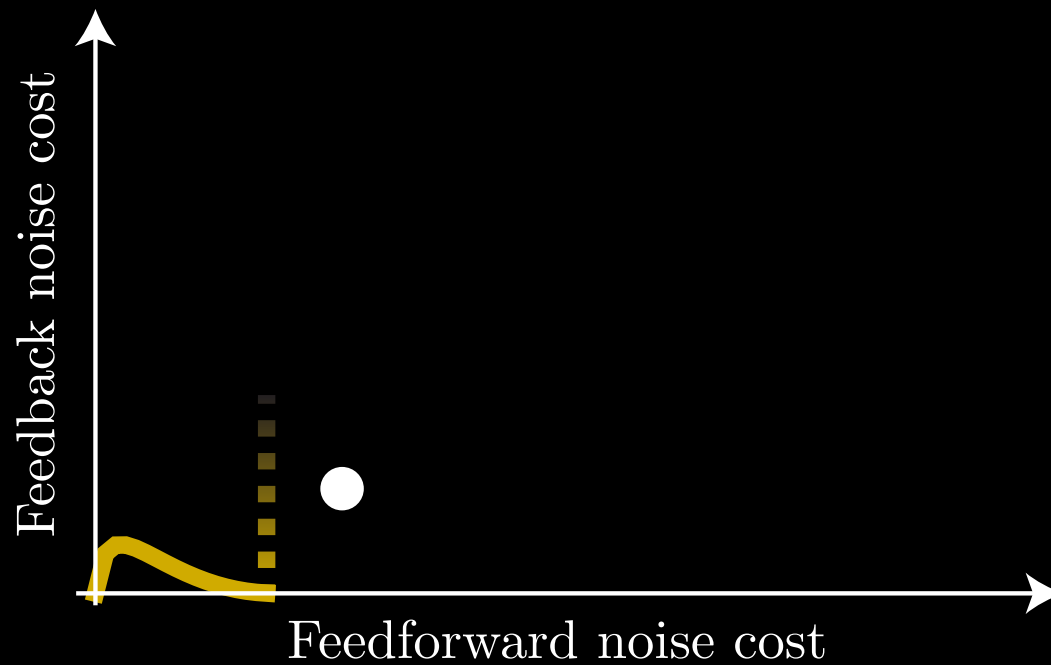
The best strategy for fixed channels (white dot)
depends on predictability.

- ↑ World time scale
- ↑ Measurement gain
- ↓ Observation noise



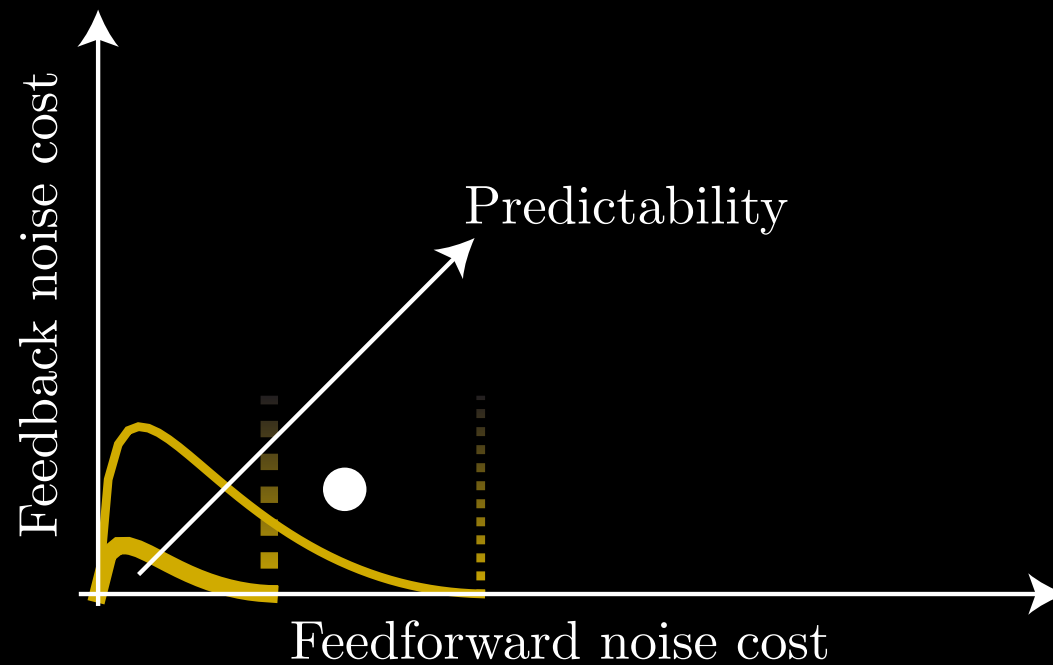
No messages sent when predictability is low

No Messages → Efficient Coding → Predictive Coding



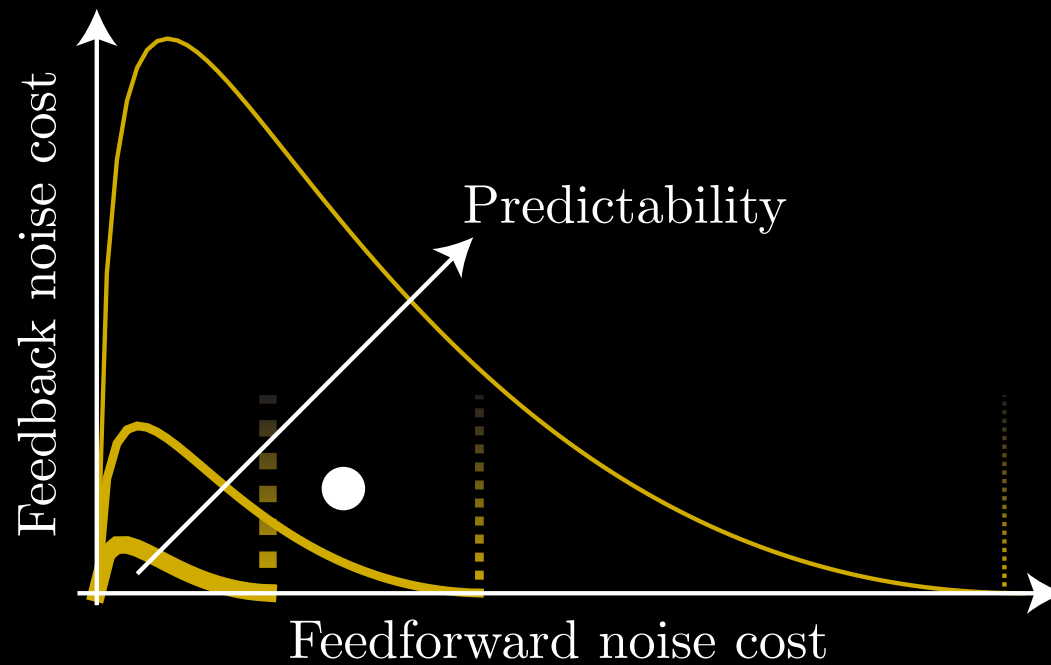
Only feedforward when predictability is moderate

No Messages → Efficient Coding → Predictive Coding



Predictive coding when predictability is high

No Messages → Efficient Coding → Predictive Coding



Outline

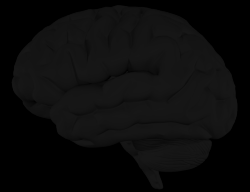
Problem
formulation



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Biological
relevance



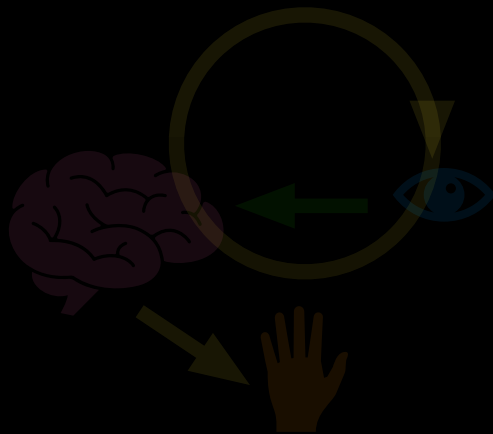
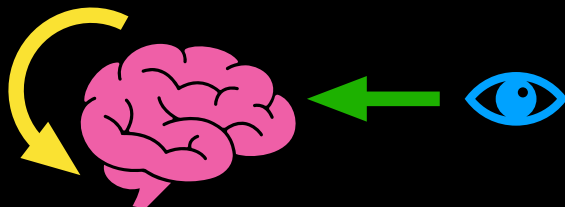
Problem
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Solution &
interpretation



Toy problems



Outline

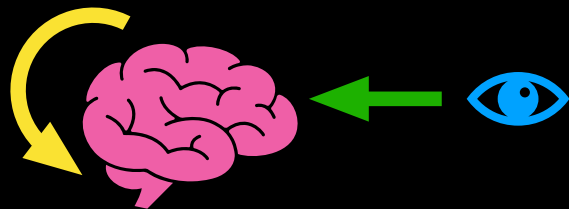
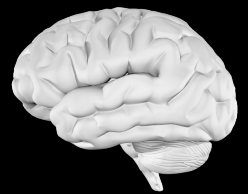
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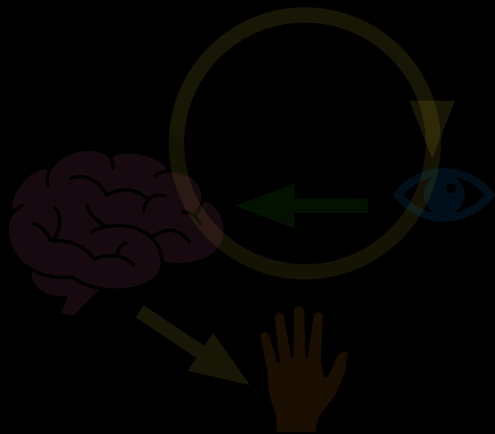
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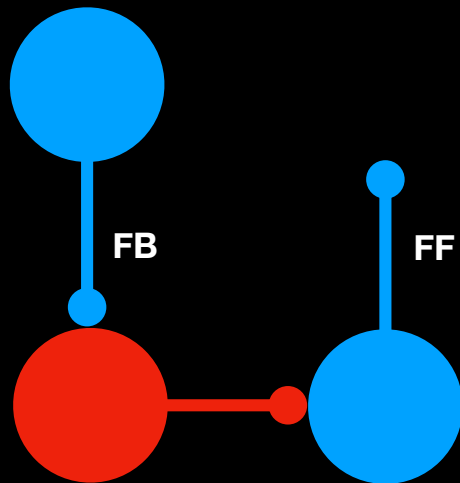
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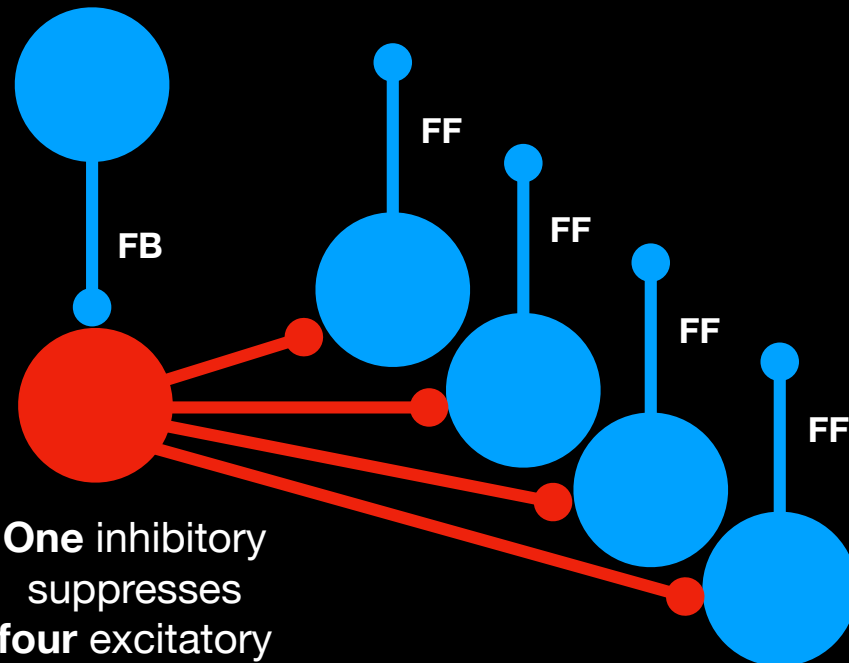
Toy problems



Why pay for inhibition to save excitation?

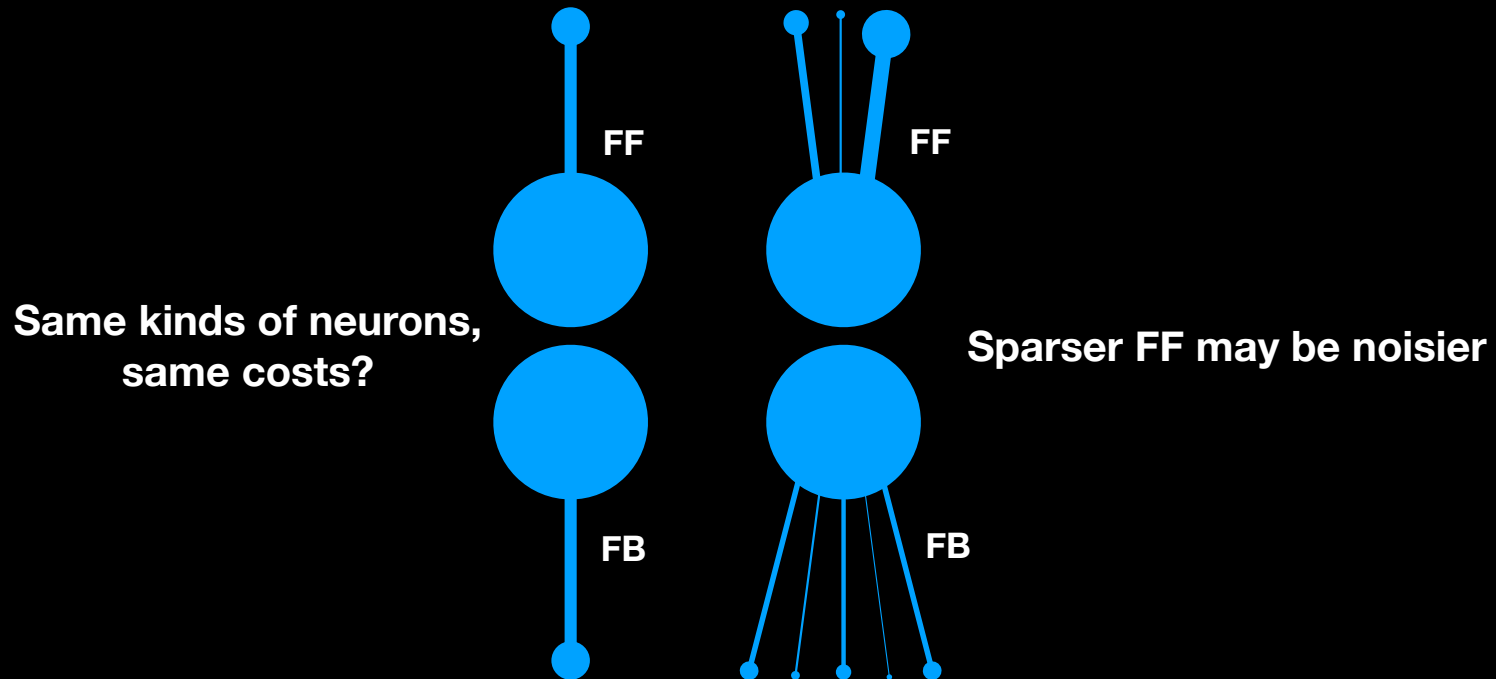


One inhibitory
suppresses
one excitatory



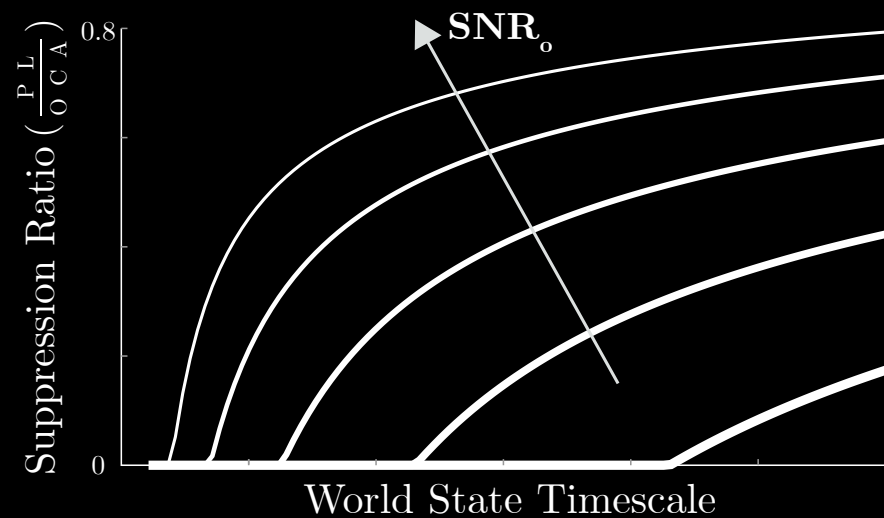
One inhibitory
suppresses
four excitatory

Why would feedback have higher noise cost?



Testable biological parameters

Context-dependent: Manipulate stimulus experimentally. E.g. time constant/predictability.



Interim conclusion

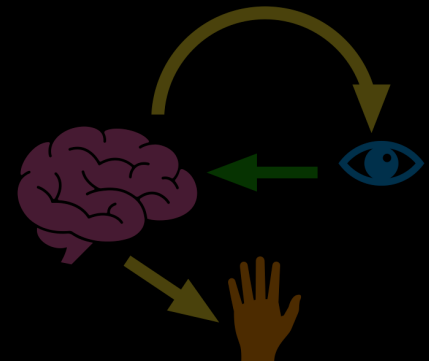
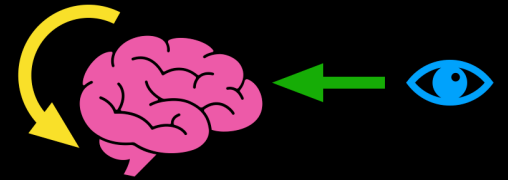
- We added biological feedback costs to predictive coding theory.
- Using feedback may be worse depending on biological constraints, computational costs, and world dynamics

Published in Advances in Neural Information Processing Systems (2022).

Recap

Brain computations under noise and energy constraints?

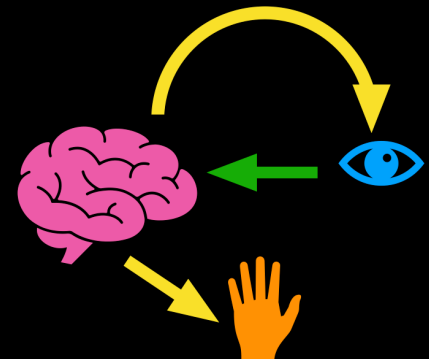
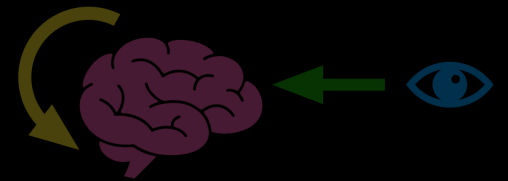
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Next

Brain computations under noise and energy constraints?

- Part I: Inference under resource constraints
 - An **abstract** mathematical model to study how brain controls **internal feedback**
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Attention is a resource



- How does the brain trade off attention costs against its benefits?
- Study this for the case of a mice performing an auditory feature based attention task.

Attention task with different task utilities

- Mice must attend to auditory features to obtain rewards.
- Varying incentives to attend - low and high juice rewards
- Pupil size to measure arousal.

Trial live audio

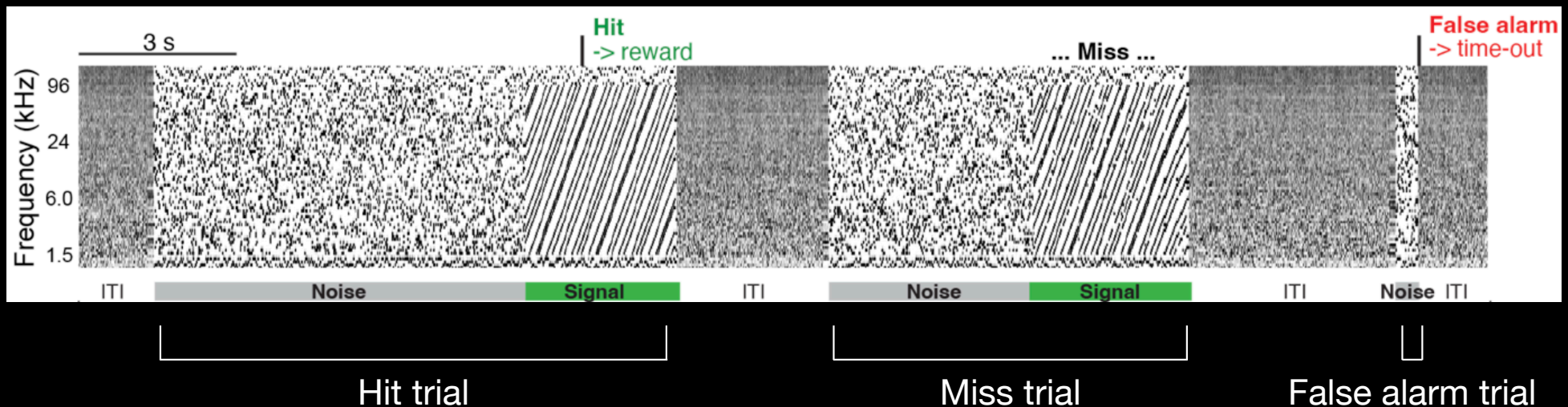


Image from de Gee, J. W., et al. "Strategic self-control of arousal boosts sustained attention." bioRxiv (2022): 2022-03.

Problem

- **Understanding mouse's computations in trading off reducing internal attention costs against maximizing external food rewards.**
- **Build reinforcement learning models to mimic mouse behavior, analyze their policies, and reverse-engineer the various cost parameters.**

Outline

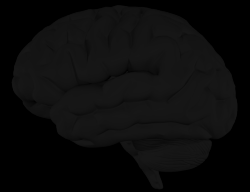
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formulation



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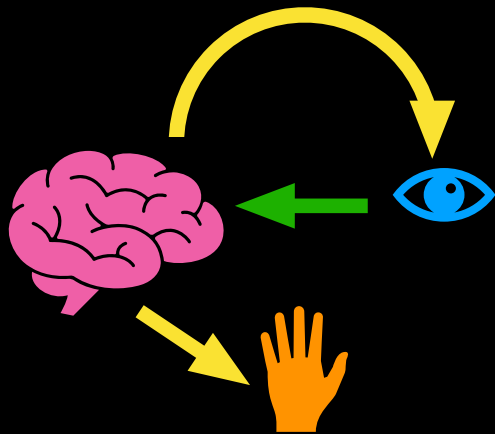
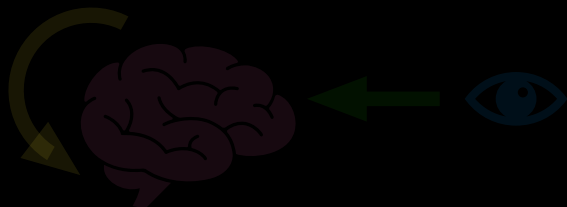
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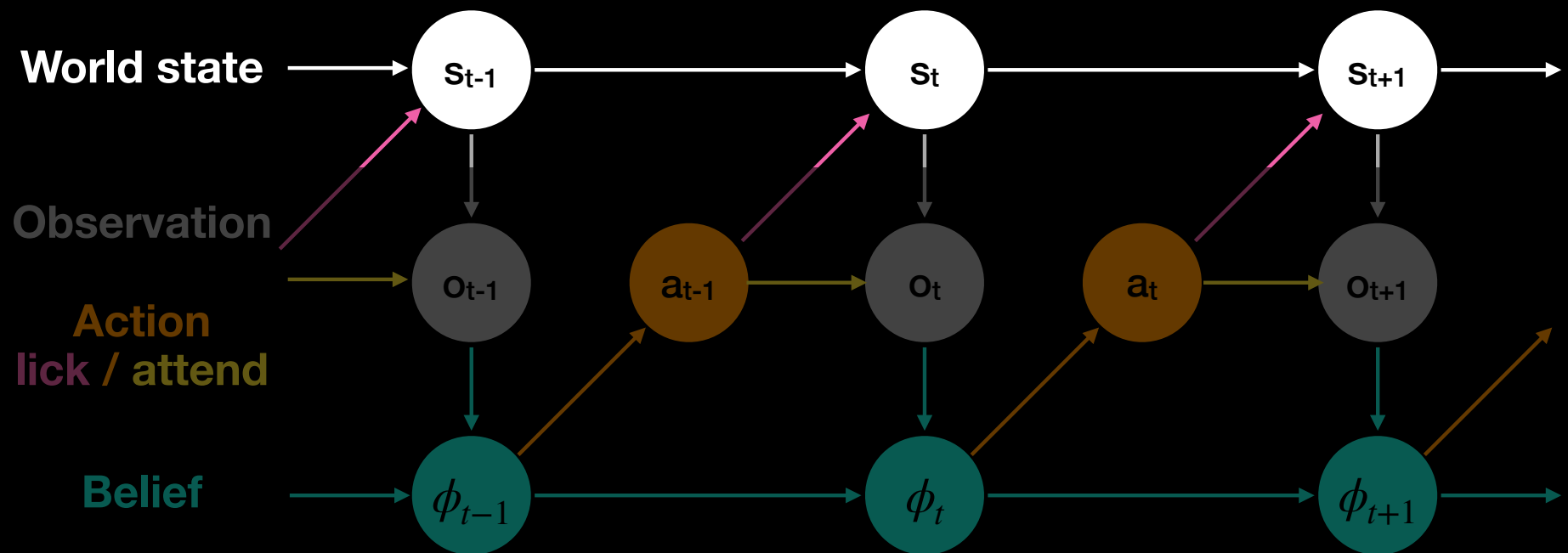
Solution &
interpretation



Toy problem



Partially observable Markov decision process

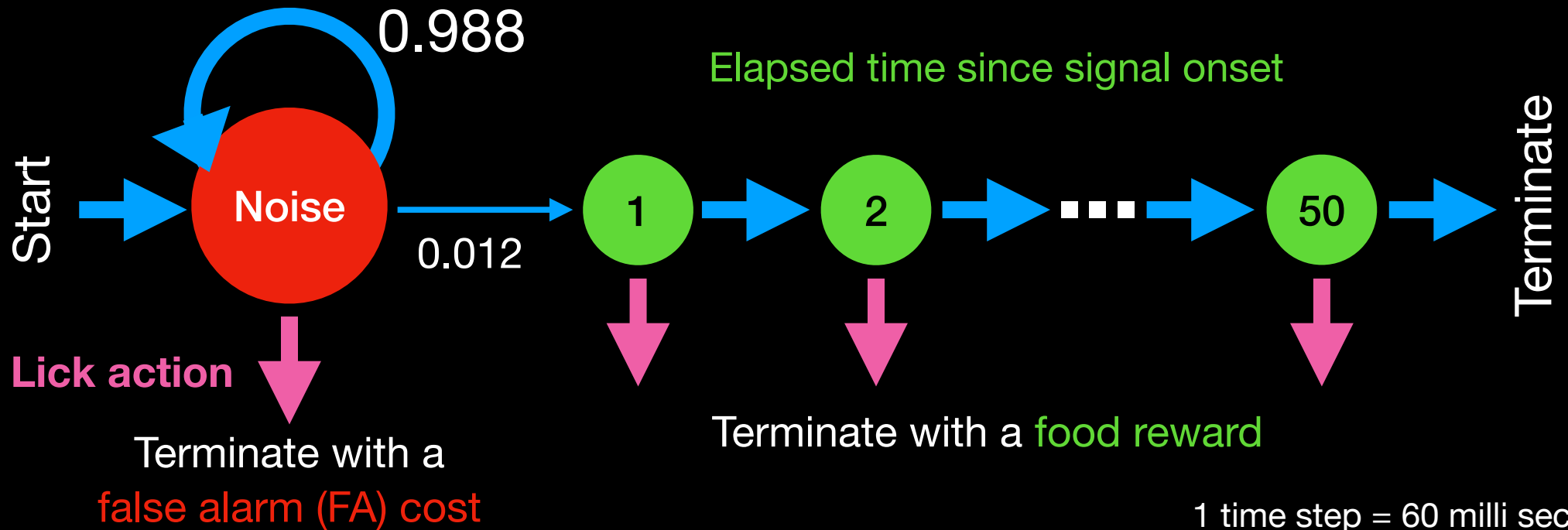


Single trial model - state transition

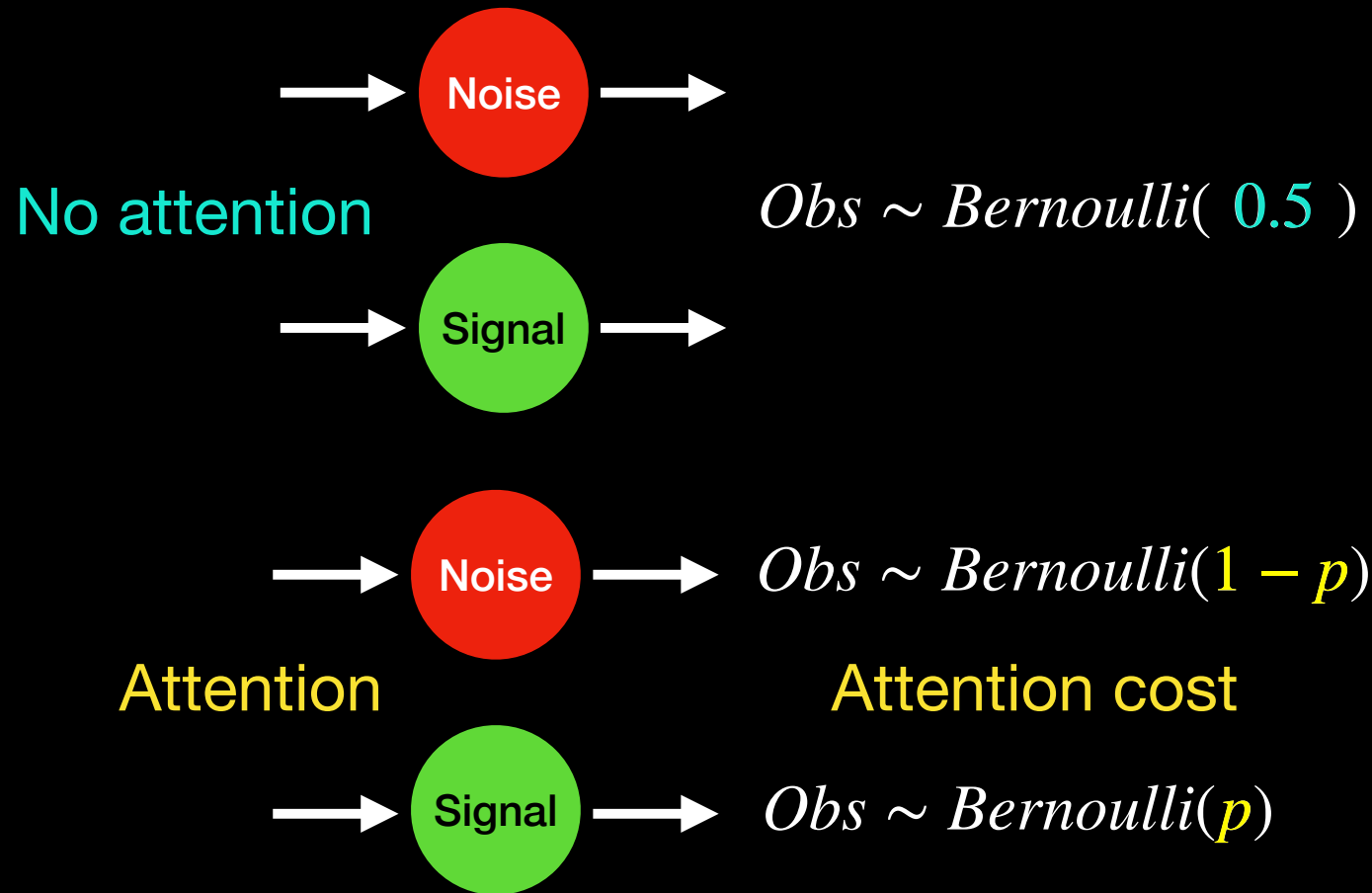
Mean 5 secs in noise

3 secs in signal

No-lick action

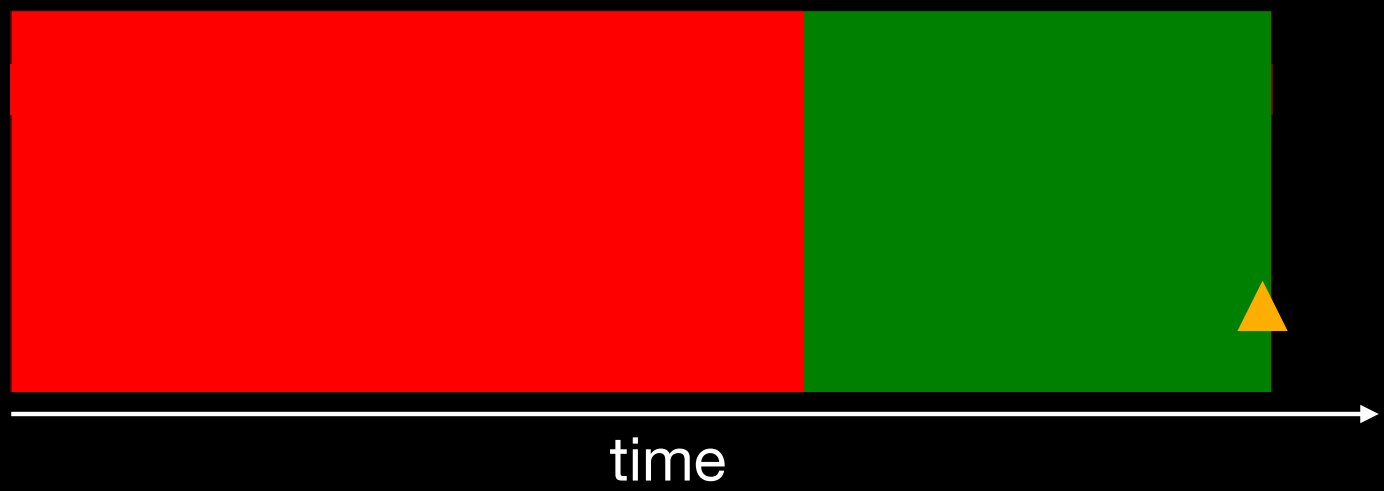


Single trial model - observation



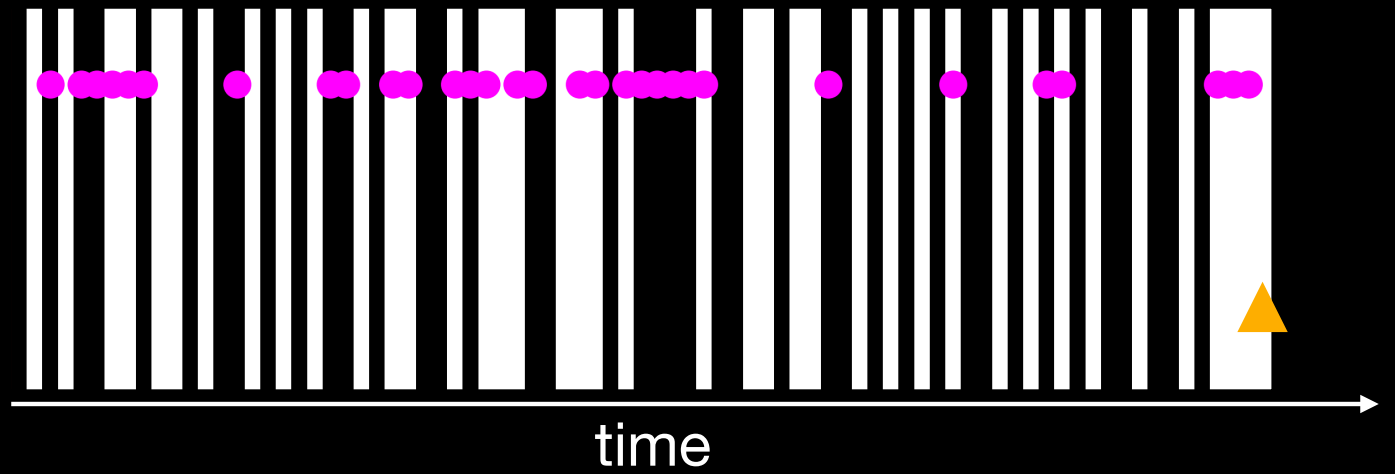
Experimentalist's view:

noise
signal
▲ lick



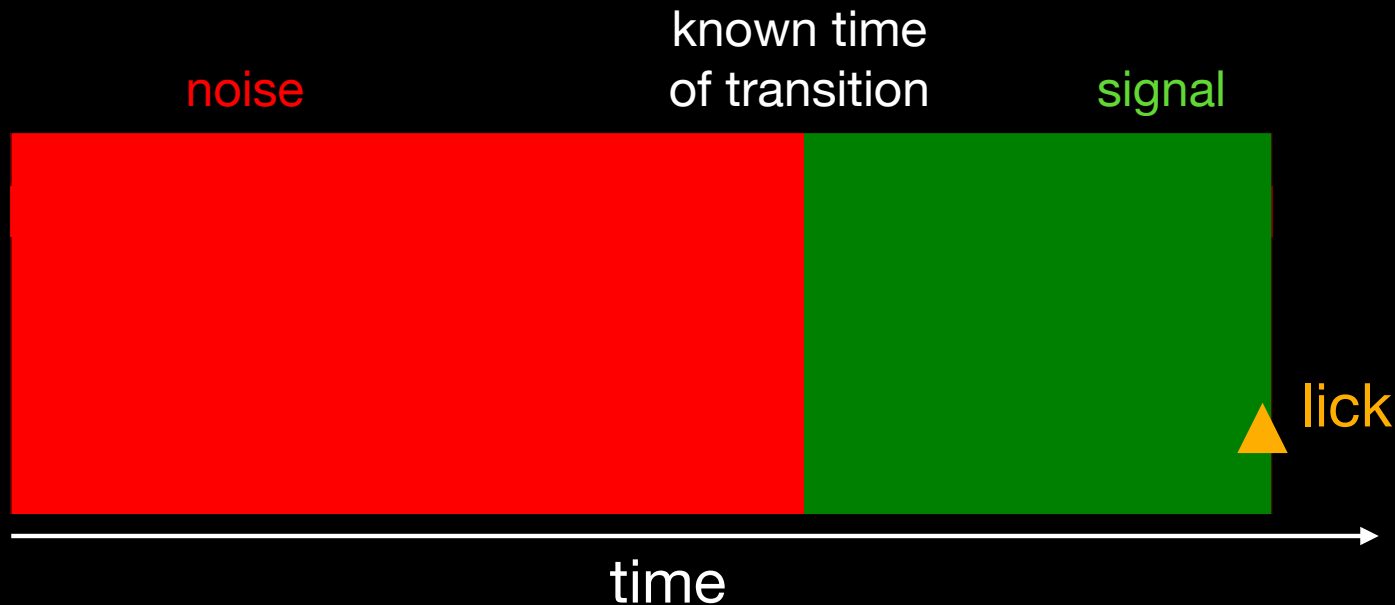
Mice's view:

□ obs 0
■ obs 1
● attend
▲ lick



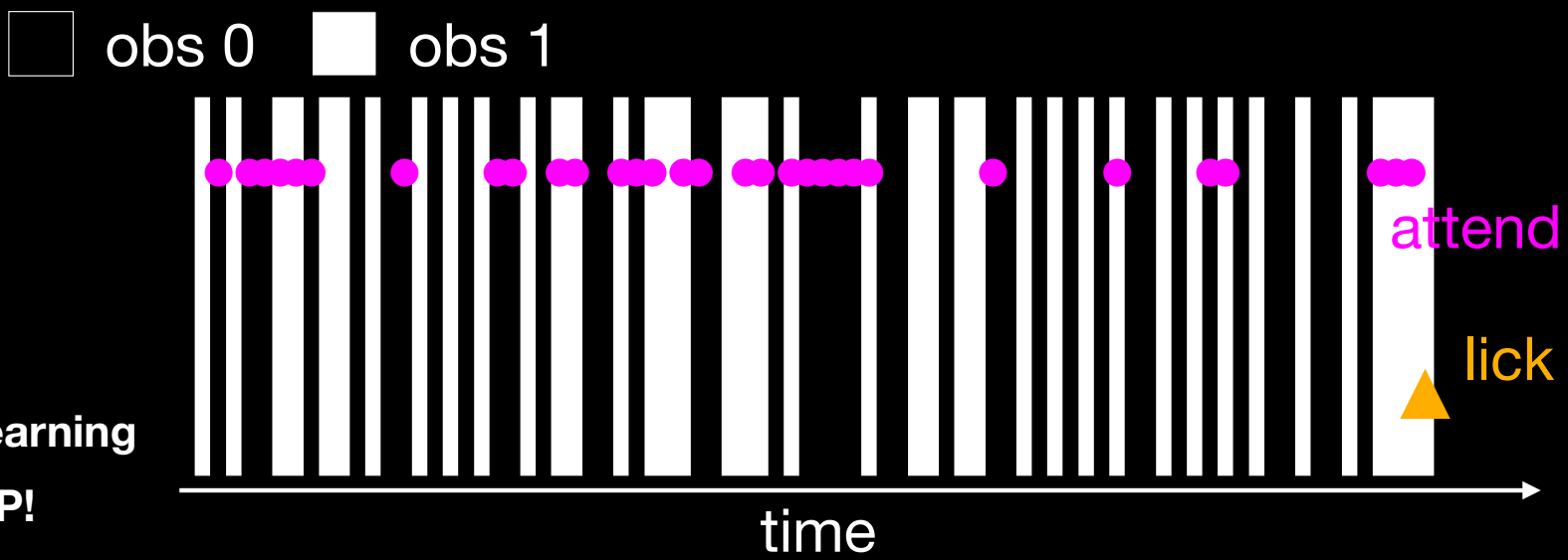
Experimentalist's view:

- Unknown:
- attention choices
 - observations
 - beliefs



Mice's view:

- Unknown:
- true state observed partially if attended to



**Apply Reinforcement Learning
to solve the POMDP!**

Outline

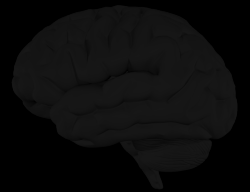
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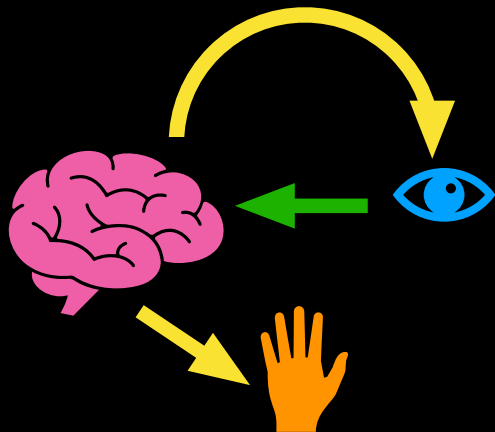
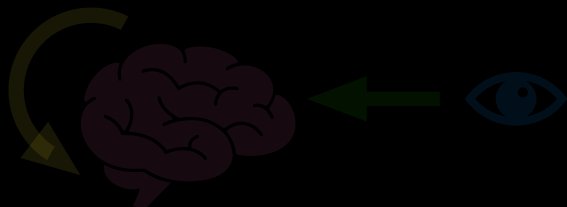
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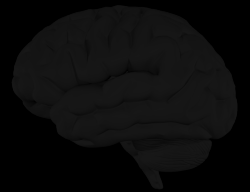
Problem
formulation



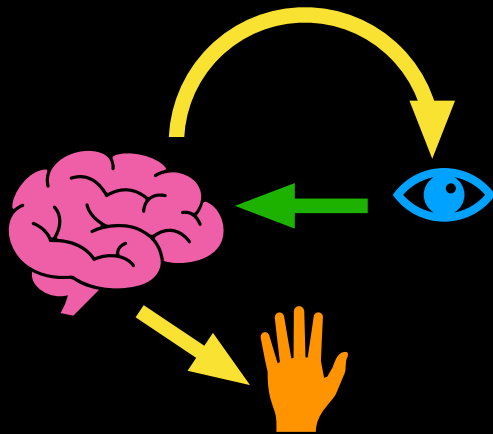
Solution &
interpretation



Biological
relevance



Problem
formulation



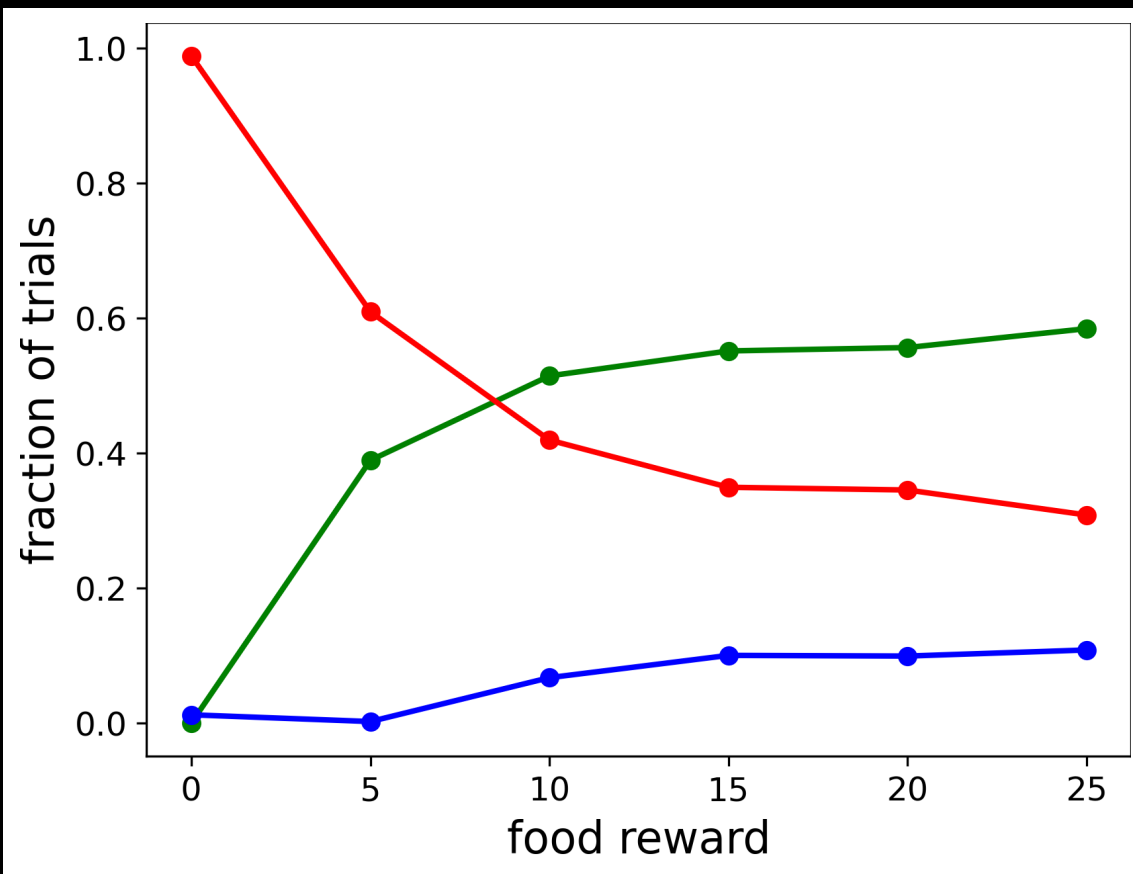
Solution &
interpretation



Toy problem



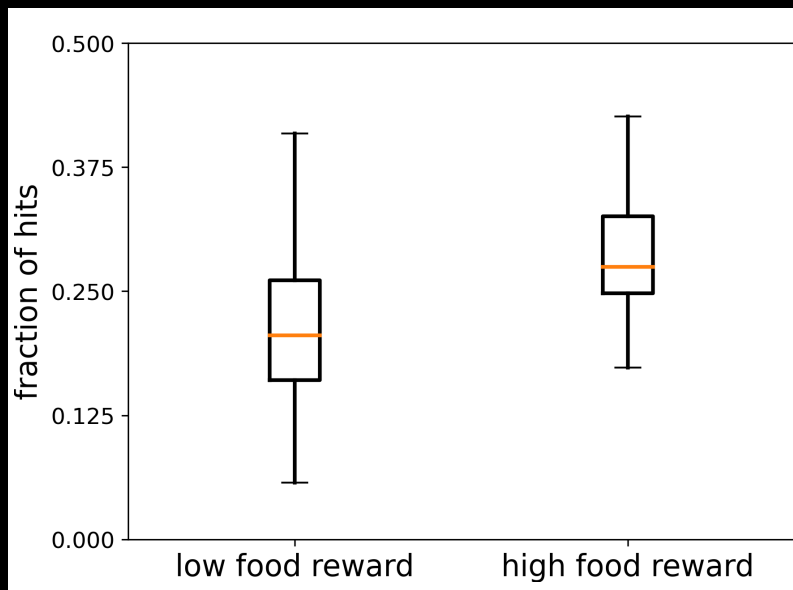
Behavior without penalty for false alarm



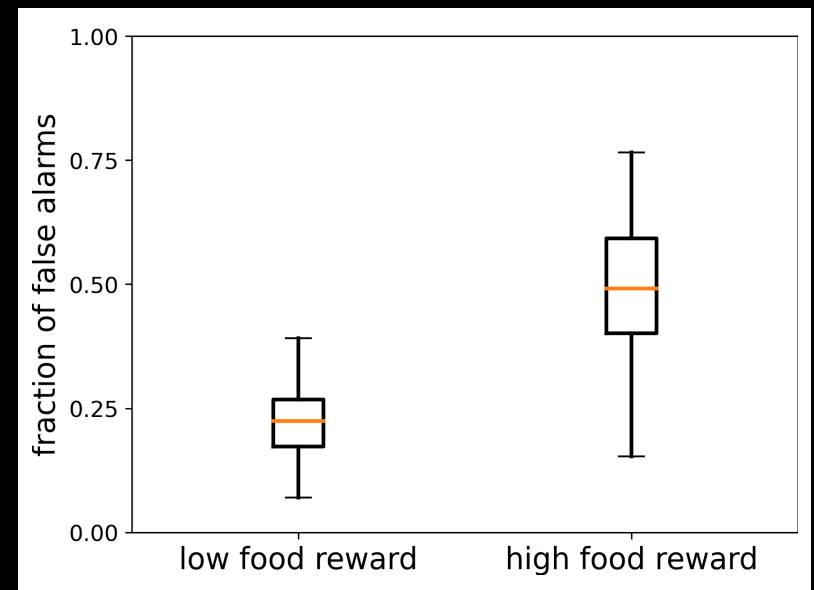
- **Hit:** lick during signal.
- **False alarm:** lick during noise.
- **Miss:** Didn't lick at all.

Interesting behavior in dataset

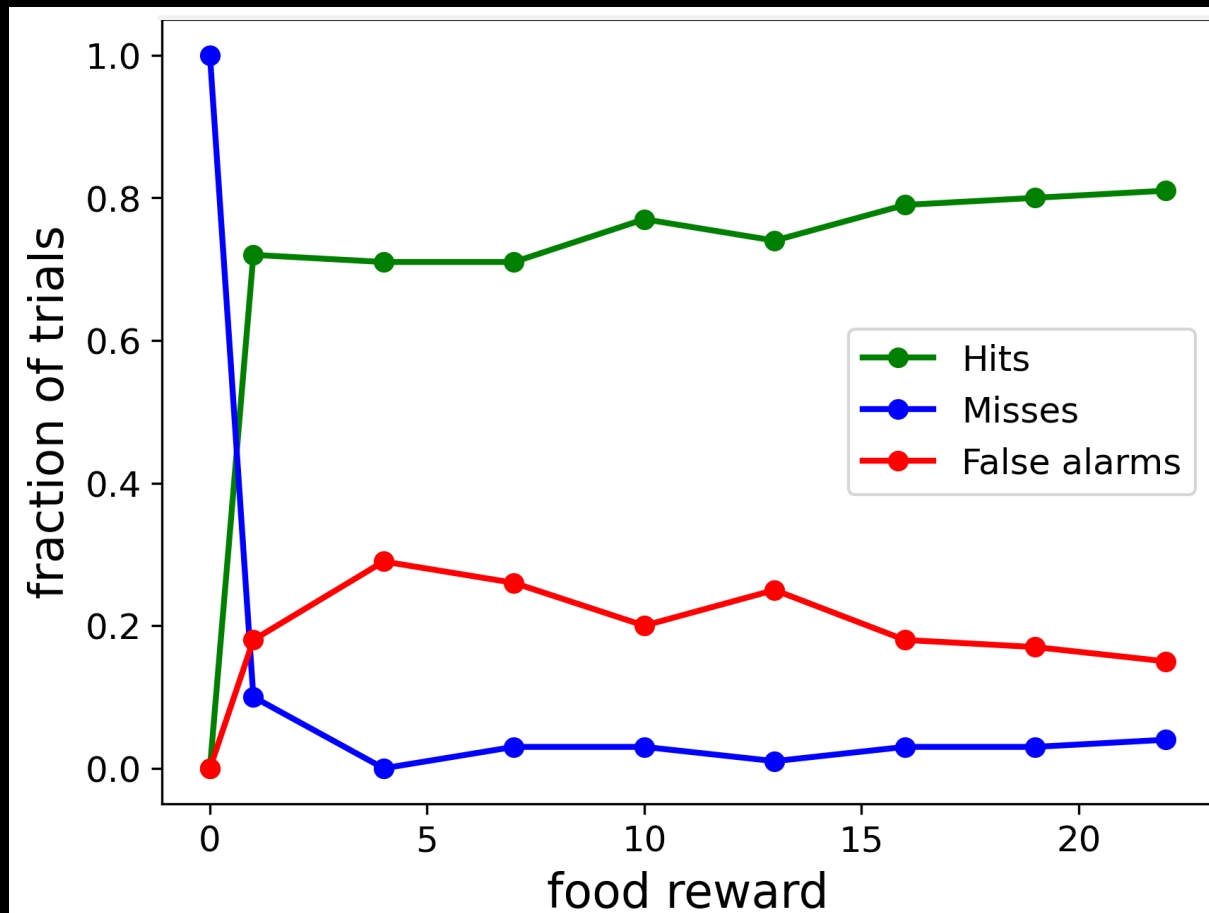
Hits increase with food reward



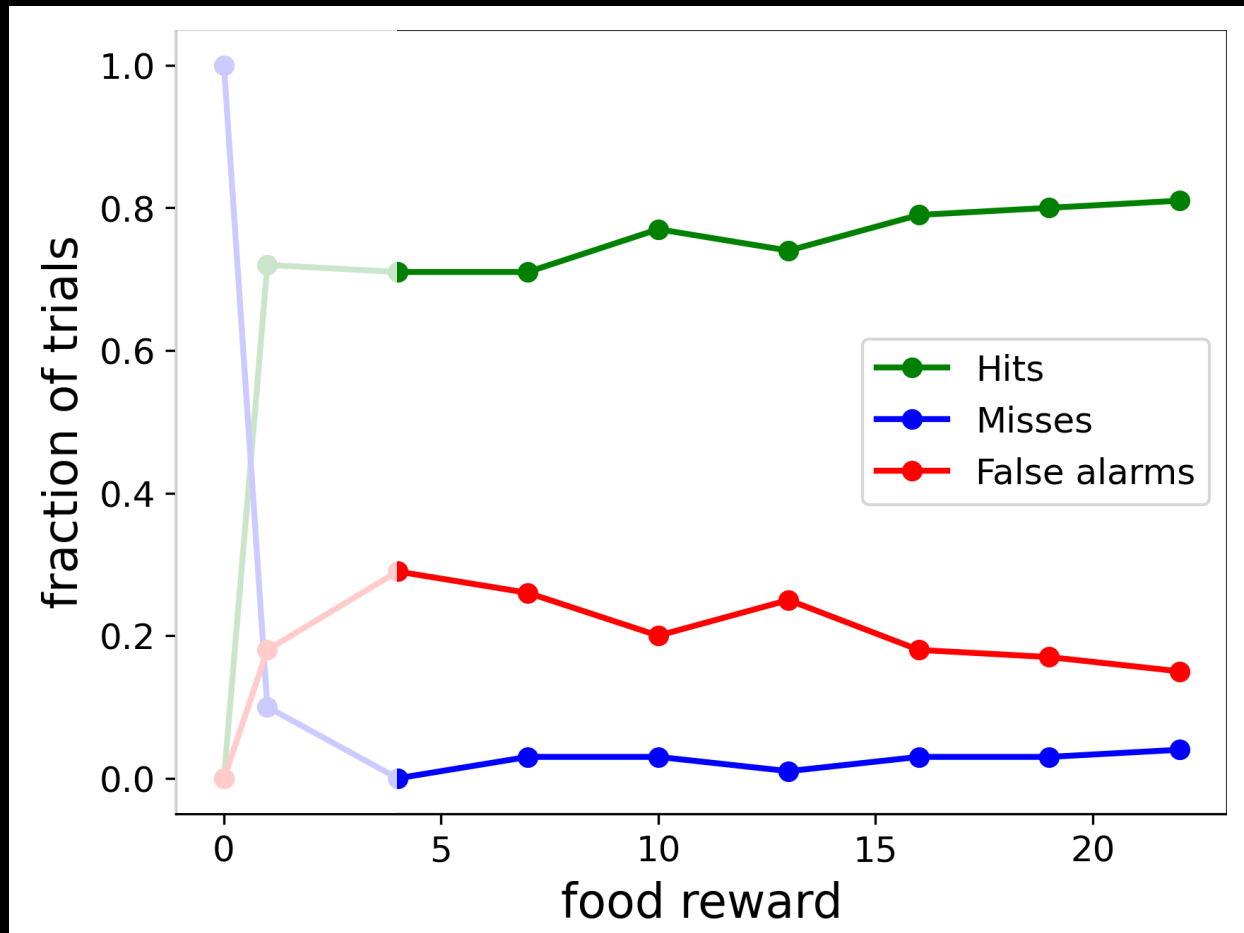
FAs increase with food reward



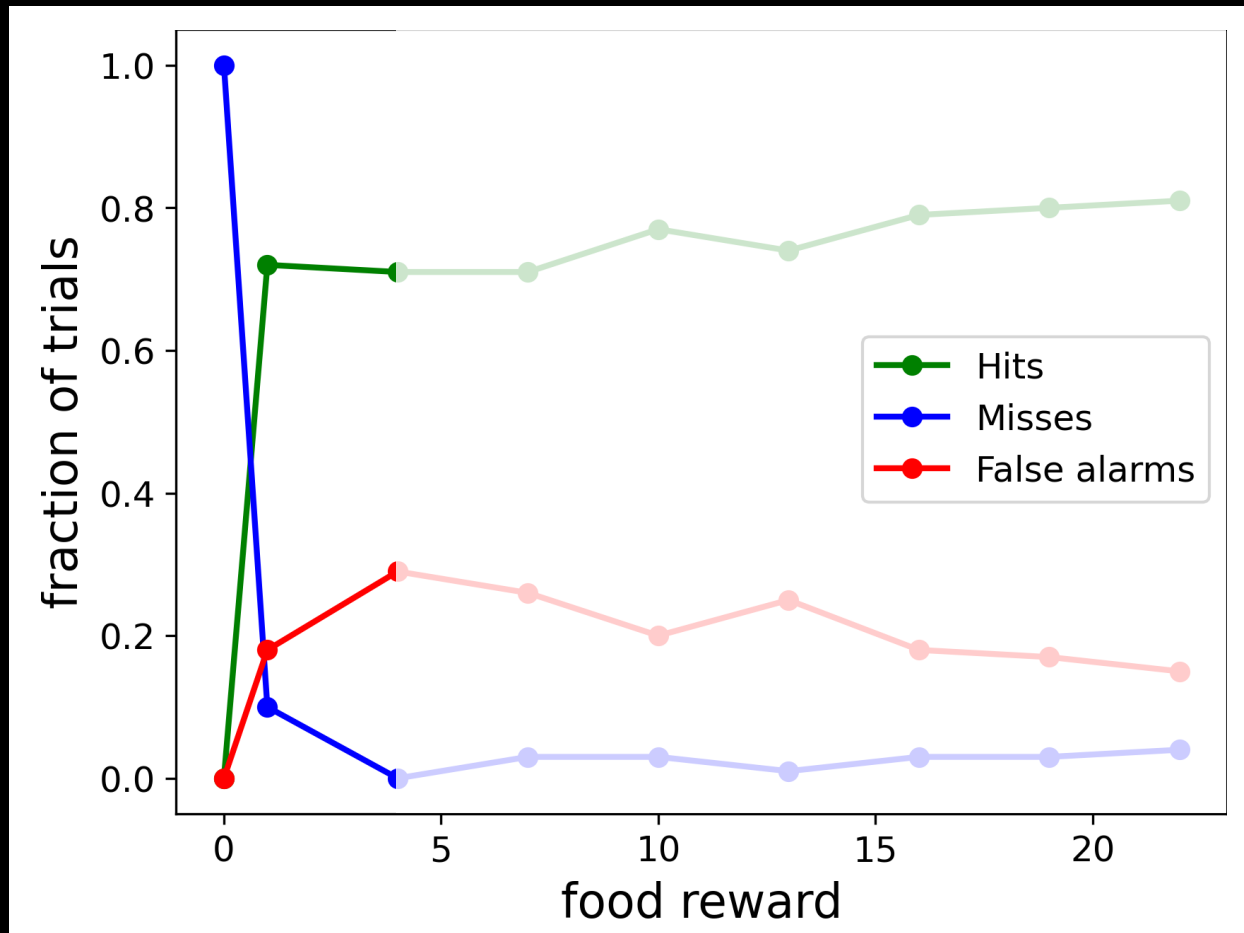
Behavior with penalty for false alarm



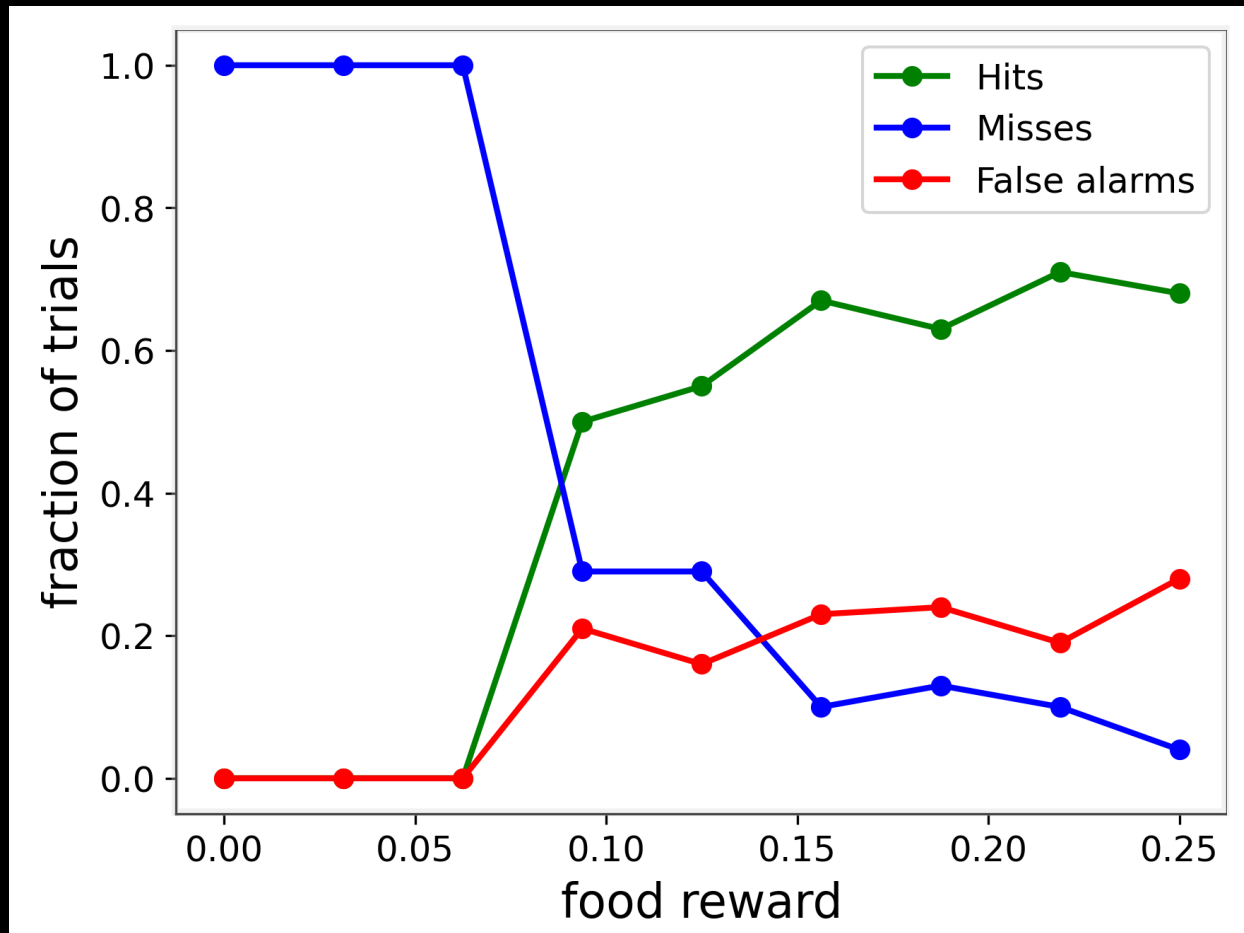
Higher food reward, attend more



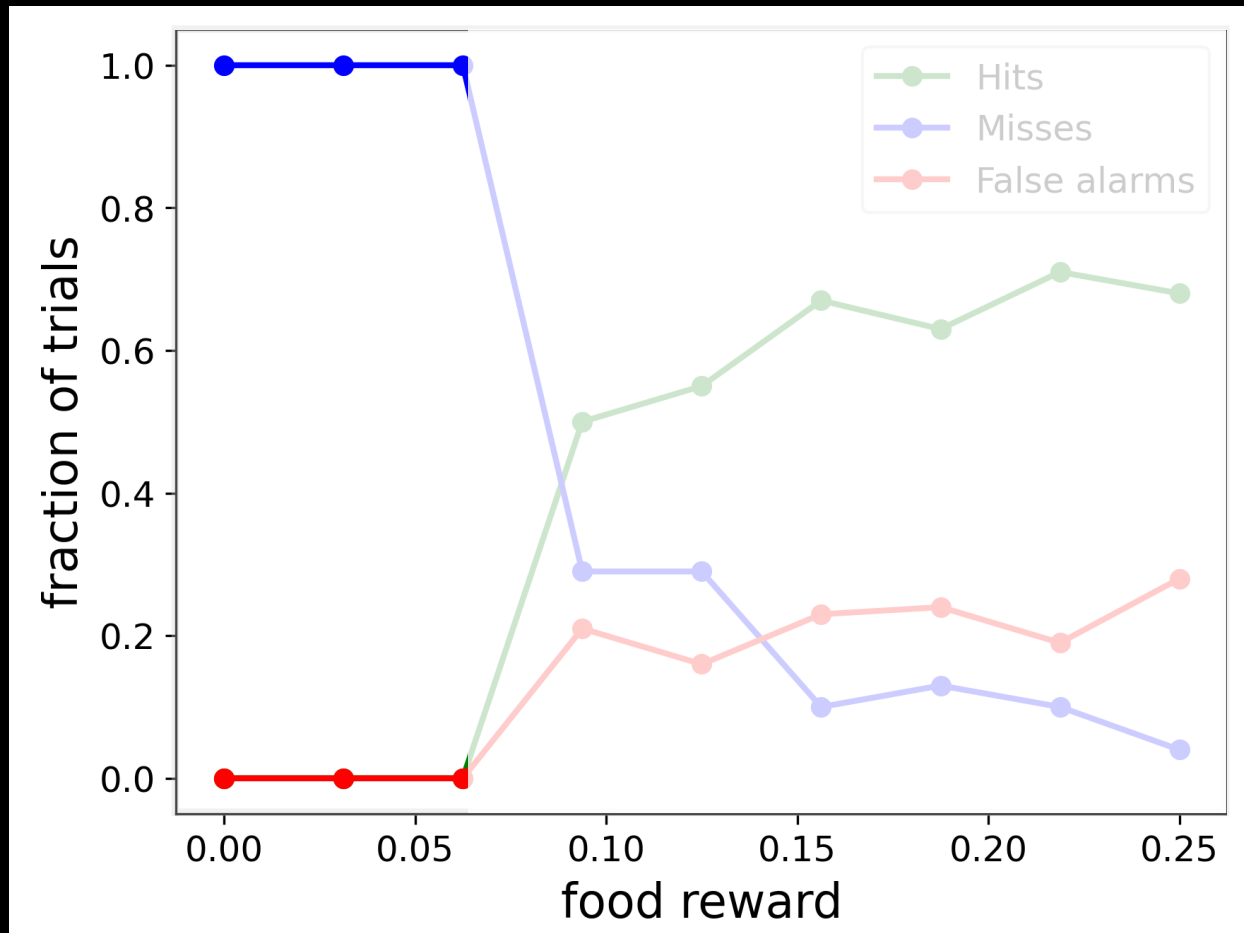
False alarms increasing with food reward



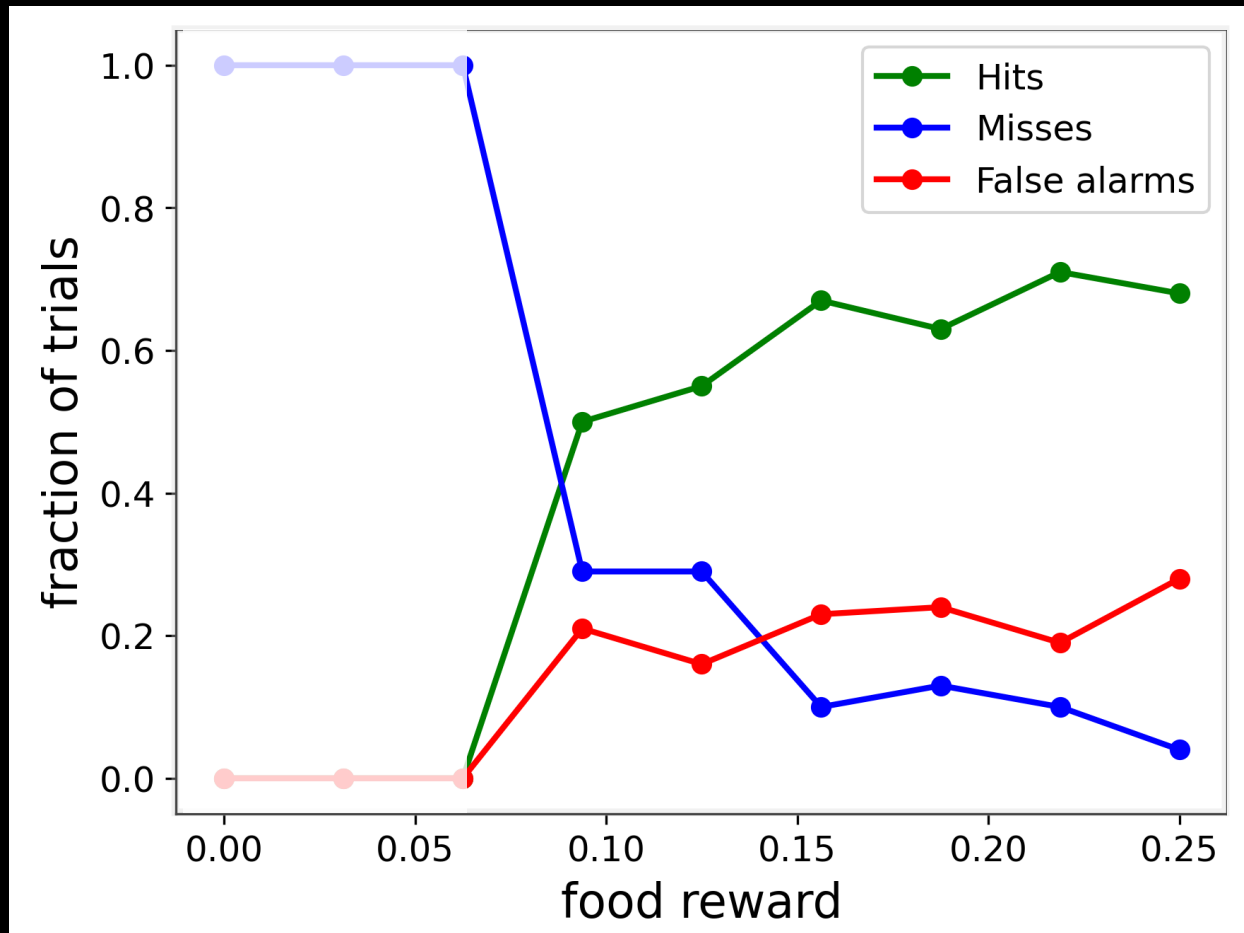
Higher food reward, higher false alarms



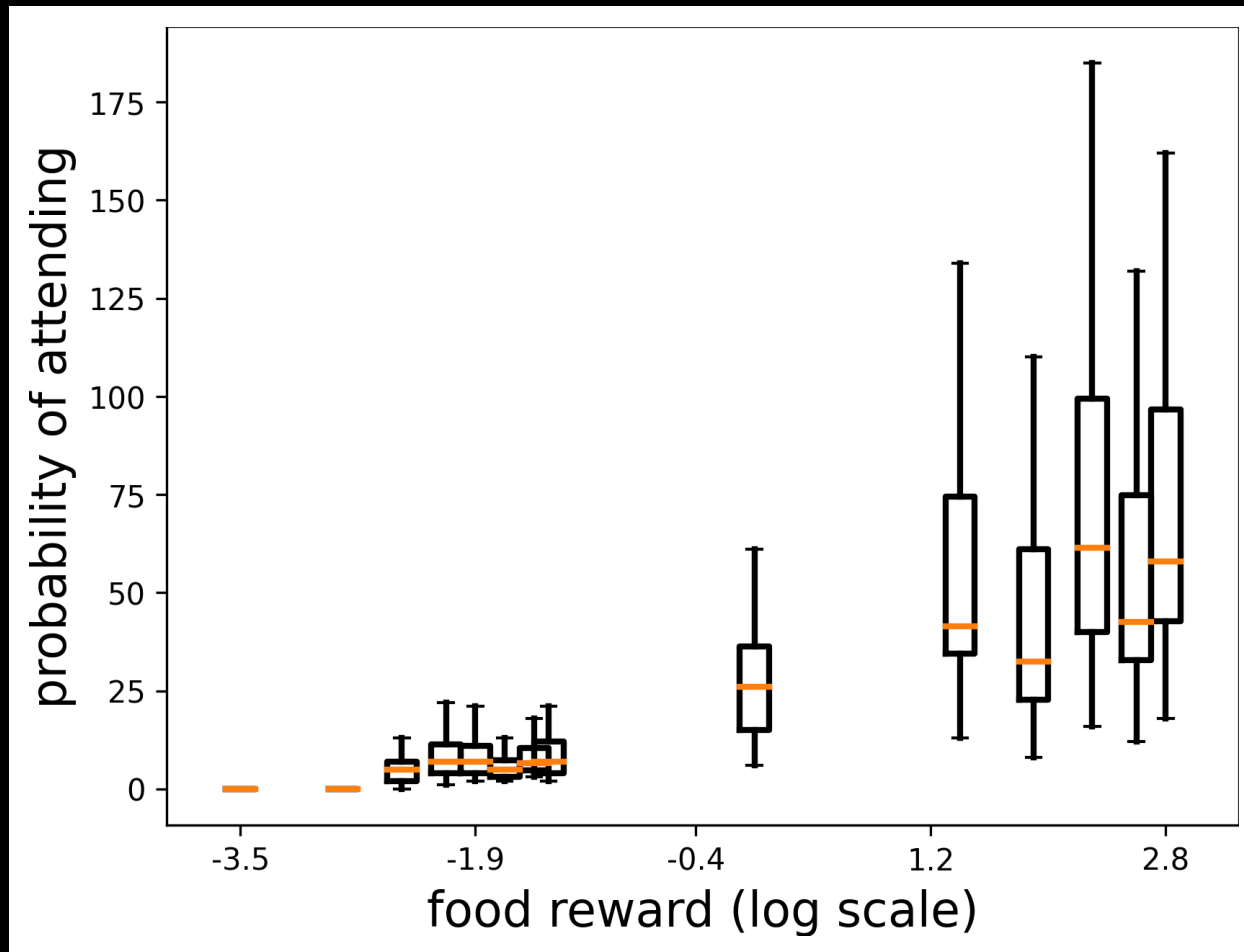
False alarm penalty too high to lick at all



Higher food reward, lick more



Increasing attention with food reward



Outline

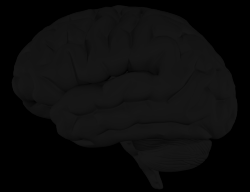
Problem
formulation



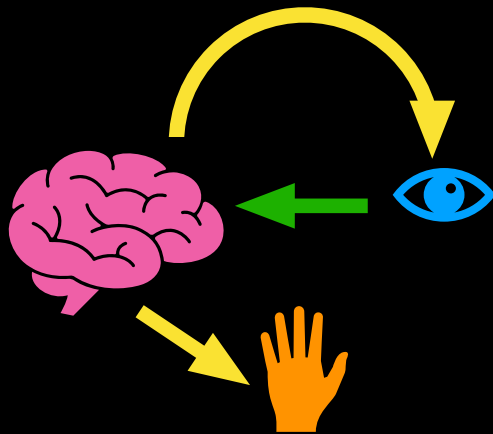
Solution &
interpretation



Biological
relevance



Problem
formulation



Solution &
interpretation



Toy problem



Outline

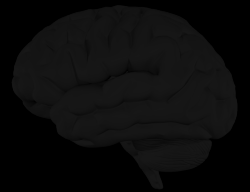
Problem
formulation



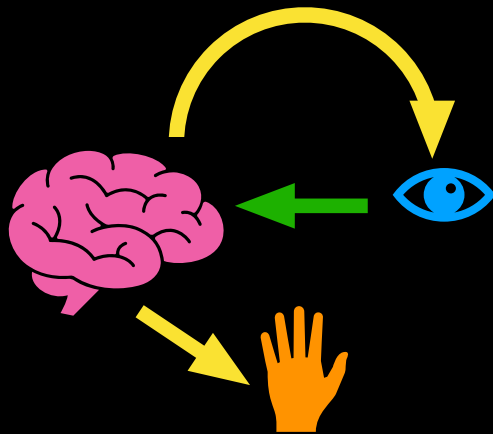
Solution &
interpretation



Biological
relevance



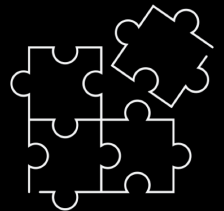
Problem
formulation



Solution &
interpretation



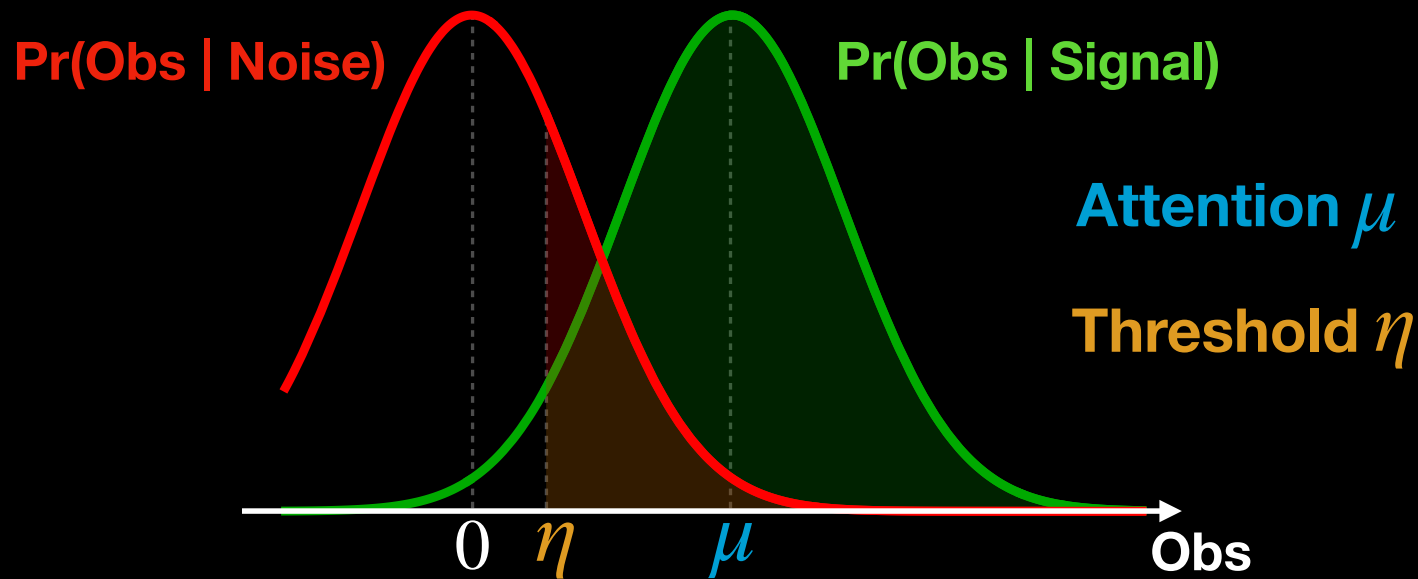
Toy problem



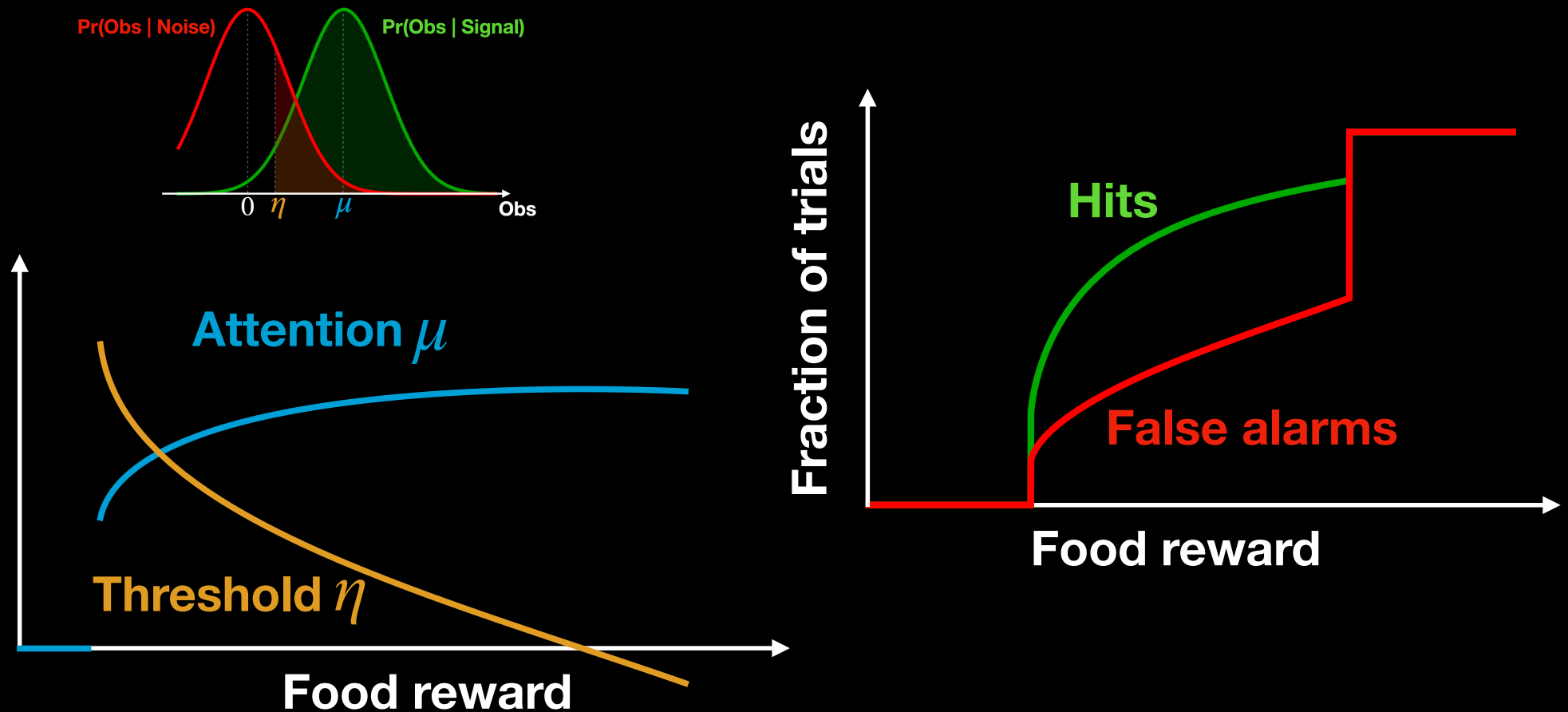
Capturing attention cost

Total reward = food reward + false alarm cost + attention cost

$$r_{total} = 0.5 r_{food} Q(\eta - \mu) + 0.5 r_{FA} Q(\eta) - \mu^2; \mu, r_{food} \geq 0, r_{FA} \leq 0$$



False alarms & hits increasing with food reward



Interim conclusion

- Model showed regulation of attention with reward size.
- Model mimicked the behavior of false alarms increase with reward size.
- Toy problem revealed responses increase with reward size.

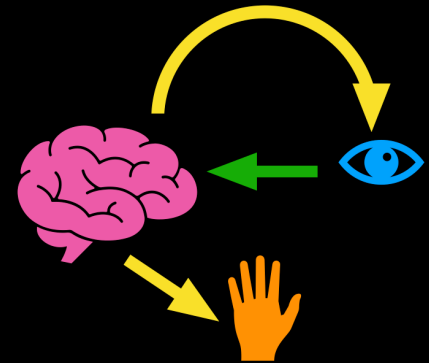
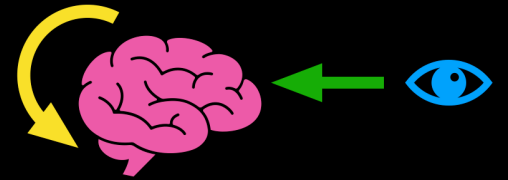
Next steps

- Analyze toy problems to build intuitions on when attention is useful.
- Comparing model vs data; performance, pupil ...
- Predictions: Thirsty mice, heightened perceived food value -> increasing false alarms with big rewards expected.

Conclusion

Explored how brain performs computations under noise and energy constraints. Our theories suggest:

- Regulating internal feedback during inference depending on biological constraints, computational costs, and world dynamics.
- Regulating external feedback (attention & action) depending on task utility and attention cost.



Thank you all for paying attention